

Full Length Article

Neural dynamics and synaptic plasticity in simple networks drive Lévy flight foraging and obstacle avoidance behaviors for bio-inspired autonomous flight

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ARTICLE INFO

Keywords:

Neuro-Lorenz attractor
Lorenz attractor
Neural dynamics
Lévy flight
Foraging behavior
Adaptive exploration control
Adaptive obstacle avoidance
Synaptic scaling

ABSTRACT

Animal foraging behavior is essential for survival. This complex behavior involves a combination of local search, seeking food nearby, and global search, exploring large areas for new food resources. These strategies can be adaptively alternated. During exploration, animals also navigate obstacles. Several studies have focused on developing bio-inspired exploration mechanisms for artificial agents, with Lévy flight emerging as one of the most effective methods. However, existing Lévy models rely on mathematical representations of Lévy distributions, which are difficult to directly relate to the biological neural systems of animals. From this perspective, we propose an efficient and adaptive neural exploration control system synthesized from a small recurrent neural network. This network, characterized by synaptic weights and adaptive inputs, exhibits neural dynamics similar to the discrete-time Lorenz attractor. The network's output provides the baseline input signal for post-processing through an adaptive bias thresholding mechanism, which generates exploration commands. These commands enable adaptive exploration behaviors that incorporate both local and global search strategies. Additionally, to ensure safe exploration in obstacle-filled environments, we integrated an adaptive neural obstacle avoidance control into the system. We assessed the performance of the proposed system in both open and closed environments using simulated and real flying robots. The resulting flight trajectories demonstrated animal-like foraging behaviors, including local and global search strategies similar to Lévy flight, along with adaptive obstacle avoidance capabilities. Our findings provide a deeper insight into the neurobiological basis of efficient foraging strategies and advanced robotic systems for autonomous exploration and navigation in complex environments.

1. Introduction

Biological search and exploration behaviors¹ are fundamental for the survival of animals in the natural world. Studying these behaviors offers valuable insights into an animal's strategies for acquiring resources, navigating its environment, and adapting to changing surroundings. Biologists and ecologists have conducted extensive research on the movement patterns of various species to better understand the underlying mechanisms that govern these behaviors (Giuggioli & Bartumeus, 2010; Yong, 2020). The studies began with the analysis of flying trajectory data from albatrosses, which roam over the open ocean in search of food. The resulting data revealed two distinct behaviors: (1) when a tar-

get is detected within range, the animal moves in a straight line toward it, and (2) when no target is detected, the animal chooses a random direction (θ) and a distance (l_x) based on the Lévy flight phenomenon (Viswanathan et al., 1996). Lévy flight is mathematically characterized by a probability distribution with a power-law tail: $P(l_x) \sim l_x^{-\alpha}$, where $1 < \alpha \leq 3$, l_x denotes the flying or moving step length (distance), and α represents the power-law (Lévy) exponent (Viswanathan et al., 1999). This distribution indicates a scale-free, long-range movement pattern.

In principle, Lévy flight is associated with search and exploration behaviors, particularly in the context of animals foraging for food. These complex behaviors result from the interplay of two distinct strategies (Reynolds & Rhodes, 2009). The first is local search, where animals

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¹ The difference between search and exploration behaviors is that search behavior involves goal-oriented movement, such as finding or looking for specific targets, whereas exploration behavior entails moving to discover available resources in an open-ended manner.

actively search for and collect nearby food resources. This strategy relies on both the intrinsic neural system (Clement et al., 2023; Deeti et al., 2023) and extrinsic sensory information, such as visual or olfactory processing systems (Buehlmann et al., 2020; Sun et al., 2020, 2021), to efficiently locate and gather food or prey. The second strategy is global search, where animals traverse larger areas to explore new food sources. This strategy is typically employed when extrinsic sensory information is limited, as often occurs in resource-scarce environments (Humphries et al., 2012). In such cases, the intrinsic neural system serves as an internal drive, generating search and exploration behaviors in animals (Sim et al., 2019) and enabling long-large distance movement to explore new resource areas.

To understand and replicate efficient search and exploration behaviors modeled by Lévy flight, several robotic studies have integrated this model into environmental exploration tasks, using it as an internal-pattern-generating mechanism to reproduce animal exploration behavior (Aaron et al., 2009; Abe, 2020; Pyke, 2015; Reynolds & Rhodes, 2009; Viswanathan et al., 2001). Lévy flight has been employed to guide robots in collaboratively exploring large environments for target detection, alongside traditional obstacle avoidance control, which reactively avoids obstacles during exploration (Katada et al., 2022; Katada & Yamashita, 2022; Zedadra et al., 2018). Furthermore, integrating Lévy flight (for exploration) with bio-inspired search algorithms, such as the firefly-based search (Zedadra et al., 2022) and the beetle antennae search algorithm (Xin Xu & Shen, 2020), enables robots to perform search and exploration tasks more efficiently. Since the Lévy flight search pattern combines local search and global search strategies, some studies have explored the transition strategy between these two strategies. Sardinha et al. (2020) used the Lévy flight as the underlying exploration system and applied adaptive mechanisms, including an artificial endocrine system-based strategy, to modify the search pattern. Another study employed a reward-based strategy to modulate the system and adjust search and exploration strategies (Alexis, 2020). Additionally, a study evaluated uncertainty factors influencing the autonomous robotic search strategies, including both intrinsic and environmental aspects (Garcia-Saura et al., 2021). However, existing studies rely on Lévy models characterized by mathematical formulas with power-law (Lévy) distributions. Although these models effectively capture movement patterns, they are difficult to relate to the biological neural systems of animals. To implement a complete search and exploration control system, these models also require the integration of additional control systems, such as obstacle avoidance, target approaching, and return-home strategies, to generate safe and effective search and exploration behaviors for flying robots.

This study proposes an adaptive neural exploration control as an alternative neural model for generating Lévy flight behavior in autonomous flying robots (see Fig. 1). The proposed neural control consists of a synthesized recurrent neural network characterized by its synaptic weights and adaptive inputs, designed to exhibit neural dynamics similar to the discrete-time Lorenz attractor (Munmuangsaen & Srisuchinwong, 2018; Viswanath, 2003). The network output, which mimics Lorenz attractor-like neural dynamics², serves as a baseline signal for generating adaptive exploration behaviors through an adaptive bias thresholding mechanism. The adaptation of bias values within the thresholding mechanism allows the neural control to generate both local and global search strategies, closely resembling Lévy flight. Additionally, the adaptive neural obstacle avoidance control (Pedersen & Manoonpong, 2018), based on hysteresis neural dynamics and synaptic plasticity, is integrated to facilitate obstacle avoidance during exploration. In this study, the proposed integrated neural system was imple-

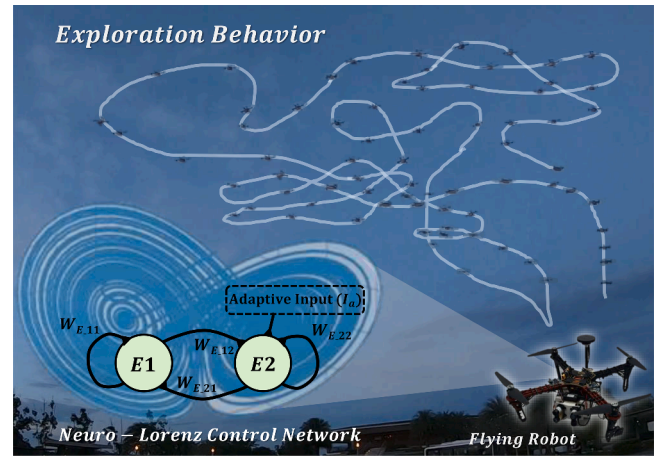


Fig. 1. The graphical abstract highlights the main contribution of this work by illustrating the neuro-Lorenz control network (NLN) in the form of a two-neuron recurrent network. This network functions as an internal driving system that generates a baseline signal exhibiting discrete-time Lorenz attractor-like neural dynamics (Zhang et al., 2015). The signal is then post-processed to generate exploration control commands for the flying robot, enabling complex exploration behaviors comparable to Lévy flight foraging.

mented on both simulated and real flying robots to evaluate its performance in obstacle-free and obstacle-filled environments. Furthermore, we analyzed how exploration movement influences the adaptability of obstacle avoidance. Lastly, the performance comparison to the state-of-the-art exploration control strategies was conducted. Taken together, this study makes the following significant contributions:

- A minimal yet effective exploration pattern generator based on a two-neuron recurrent network termed the “Neuro-Lorenz Control Network.” This network, functioning as an internal driving system, exhibits discrete-time Lorenz attractor-like neural dynamics and operates in conjunction with an adaptive bias thresholding mechanism to generate complex exploration behaviors akin to Lévy flight foraging behaviors.
- An enhanced adaptive neural obstacle avoidance control, referred to as the “Neuro-Hysteresis Control Network.” This neural control builds upon a previous version (Pedersen & Manoonpong, 2018) that utilized only front-facing sensors. The enhancement incorporates both frontal and lateral-surrounding sensors to prevent collisions and introduces a synaptic plasticity mechanism to improve obstacle avoidance adaptability during exploration.
- An integrated adaptive neural system combining the Neuro-Lorenz and Neuro-Hysteresis control networks. This integrated control system enables an autonomous flying robot (drone) to conduct exploration missions safely, incorporating obstacle avoidance. Moreover, it enhances overall flying distance by 69.98 %, 91.54 %, and 53.15 % within a specified timeframe to explore environments with low obstacle density, high obstacle density, and high obstacle density with deadlock-shaped obstacles, respectively.

The remainder of this article is organized as follows: Section 2 provides information on related works. Section 3 details the adaptive exploration and adaptive obstacle avoidance control systems and their integration, followed by their application in controlling search and exploration tasks in flying robots. Section 4 presents experimental results conducted in both open and closed environments. Finally, Section 5 provides the discussion and conclusion, respectively.

2. Related works

Environmental search and exploration strategies play a significant role in both natural and artificial systems, enabling agents to locate

² Note that the Lorenz attractor and Lorenz attractor-like neural dynamics mentioned later in this work specifically refer to the discrete-time Lorenz attractor and discrete-time Lorenz attractor-like neural dynamics (Zhang et al., 2015), respectively.

resources or targets in unknown environments without a specific goal location. One of the simplest strategies is the random walk, where an agent moves in successive steps with randomly selected directions and step lengths. A specific type of random walk, in which each movement is small and drawn from a normal (Gaussian) distribution centered around zero, is known as a Gaussian random walk (Kegeleirs et al., 2019), while its continuous-time equivalent is referred to as Brownian motion (Angstmann et al., 2019; Mun, 2017). This method has been widely employed in modeling robotic exploration systems that operate without prior knowledge of the environment or advanced detection capabilities (Pang et al., 2021; Sakamoto et al., 2024). Although easy to implement, random walks are often inefficient in covering large areas or locating sparsely distributed targets due to redundant and aimless local movements. To address these limitations, Lévy flight search strategies have been proposed and observed in both biological and artificial systems. In a Lévy flight, an agent performs many short steps interspersed with occasional long jumps, where the step lengths follow a power-law Lévy distribution (Viswanathan et al., 1996, 1999). This allows for a balance between local exploitation and global (distant) exploration, making it highly effective in environments where resources are rare and unpredictably distributed (Reynolds & Rhodes, 2009). In robotics, several studies (Katada et al., 2022; Katada & Yamashita, 2022; Zedadra et al., 2018) have employed Lévy flight strategies to drive collaborative robotic exploration in large environments for target detection, while integrating traditional obstacle avoidance controls to enable reactive navigation around obstacles during movement. Moreover, the integration of Lévy flight-based exploration with bio-inspired search techniques, such as the firefly-based search algorithm (Zedadra et al., 2022) and beetle antennae search algorithm (Xin Xu & Shen, 2020), has demonstrated improved performance in robotic search and exploration applications.

Another widely used strategy for environmental exploration is graph-based exploration, in which the environment is modeled as a graph

with nodes representing discrete positions and edges identifying possible paths. Agents navigate by moving along these edges, guided by shortest-path algorithms. This approach is highly effective in structured environments where connectivity information is available or can be constructed in real-time, such as indoor mapping (Lindqvist et al., 2021) or subterranean environments (Dang et al., 2020). It enables efficient path planning, avoids redundant visits, and minimizes travel distance to the goal. However, it depends on the agent's ability to detect or infer the environment's structure, often requiring high-performance detection systems (e.g., 3D LiDAR or 3D scanners) and substantial computational resources.

In addition to random walks, Brownian motion, Lévy flight, and graph-based exploration, recently a data-driven machine learning approach, deep reinforcement learning (DRL), has been proposed. DRL combines reinforcement learning algorithms with deep neural networks, enabling agents to learn complex policies that maximize cumulative rewards based on their experiences within the environment. Through trial and error, the agent updates its policy by receiving feedback from the environment in the form of rewards or penalties, allowing it to develop efficient exploration strategies tailored to specific tasks or contexts. DRL has been successfully applied to tasks such as autonomous drone navigation and exploration in complex environments (Cimurs et al., 2022; Taheri et al., 2024; Yue et al., 2019). While DRL can achieve highly adaptive and optimal behaviors, it comes with high computational costs, long training times, and requires carefully designed reward functions and training environments.

In response to these challenges, we propose a modular, online learning, and low-computation approach for search and exploration (see Fig. 2), based on an adaptive neural exploration control that enables autonomous flying robots to exhibit Lévy flight-like behaviors. The system employs a two-neuron recurrent network with adaptive inputs, designed to generate neural dynamics resembling the discrete-time Lorenz

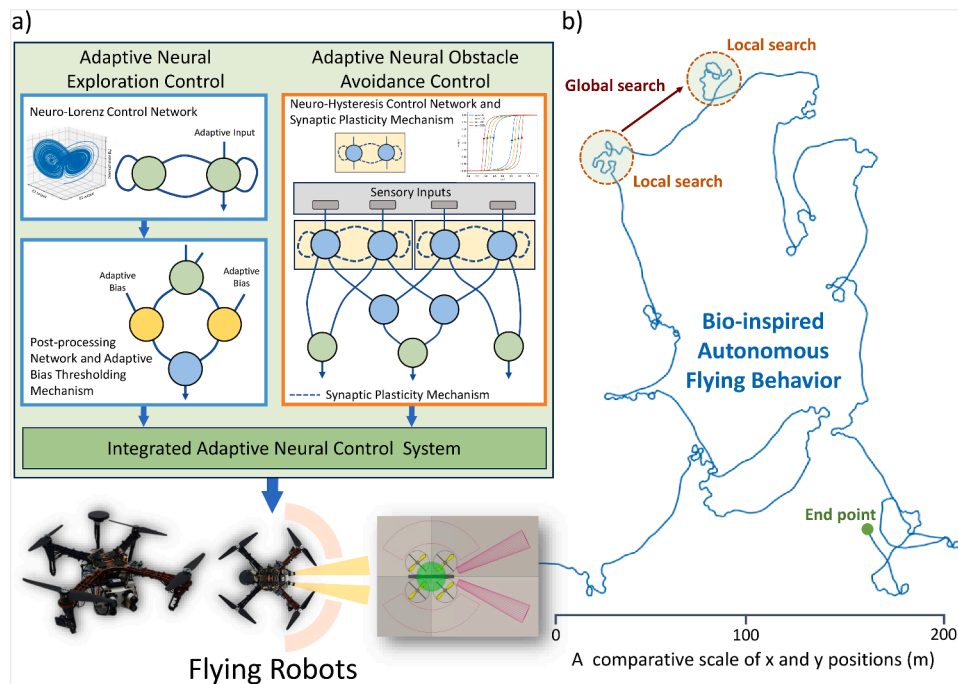


Fig. 2. (a) The overall integrated neural system for adaptive exploration and obstacle avoidance neural controls facilitates adaptive exploration and obstacle avoidance behaviors in an autonomous flying robot. This neural control system consists of two main modules: the adaptive neural exploration control (ANEC) module and adaptive neural obstacle avoidance (ANOAC) module. The ANEC module (blue edge) utilizes the neuro-Lorenz control network (NLN) along with the adaptive bias thresholding mechanism in the post-processing network to generate adaptive exploration commands. The ANOAC module (orange edge) employs two neuro-hysteresis control networks (NHCNs), which incorporate synaptic plasticity mechanisms to generate adaptive obstacle avoidance commands. The outputs of these two modules are integrated (green box) and work synergistically to generate final control commands, which are transmitted to the flying robot. (b) Tracking the trajectory when the robot is flying in an open area, exhibiting both global and local search patterns in its flight path.

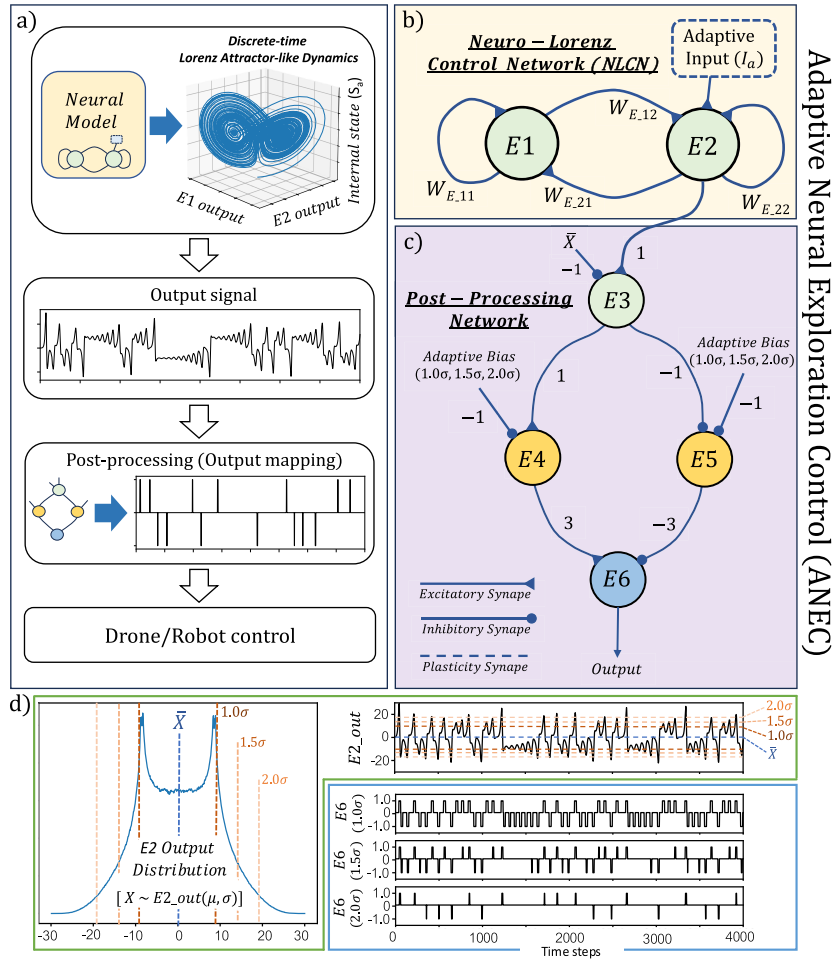


Fig. 3. (a) The overall computational process of ANEC. (b) The NLCN (consisting of $E1$ and $E2$) with (c) the post-processing network (consisting of $E3 - E6$ and an adaptive bias thresholding mechanism). The inset figure in (a) illustrates the discrete-time Lorenz attractor-like dynamics produced by the NLCN. (d) The NLCN output signal of $E2$ and its distribution (depicted by the green block). This distribution exhibits an average ($\bar{X}$) of 0.085 and standard deviation (σ) of 9.167. The post-processing output of $E6$ is utilized for activating exploration behavior with local and global search strategies under various bias values (1.0 σ , 1.5 σ , and 2.0 σ , see the blue block).

attractor. Its output is fed into an adaptive thresholding mechanism that can control local and global search strategies. Additionally, an adaptive neural obstacle avoidance control, based on neural hysteresis dynamics, synaptic plasticity, and sensory feedback, is integrated to ensure safe navigation. The system's performance was evaluated on both simulated and real flying robots in various environments, alongside analysis of how exploration patterns influence obstacle avoidance adaptability.

3. Materials and methods

This study proposes and implements an integrated adaptive neural exploration and obstacle avoidance control system designed to facilitate adaptive exploration with local and global search strategies, as well as adaptive obstacle avoidance for autonomous flying robots. The system is realized through two modular mechanisms: the adaptive neural exploration control (ANEC) and adaptive neural obstacle avoidance control (ANOAC). These mechanisms leverage neural dynamics, synaptic plasticity, and sensory feedback to govern the behavior of flying robots, as illustrated in Fig. 2.

For the ANEC, the neural dynamics representing the Lorenz attractor, embedded in the neuro-Lorenz control network (NLCN), is employed to generate “random walk” behavior, as depicted in Fig. 3. The NLCN works in collaboration with a post-processing network and adaptive bias thresholding mechanism to generate adaptive exploration behaviors, in-

cluding both local and global search strategies. For the ANOAC, the neural dynamics, representing the hysteresis effect, embedded in the neuro-hysteresis control network (NHCN), are utilized to control the turning angle for obstacle avoidance. The NHCN operates in conjunction with a synaptic plasticity mechanism, as illustrated in Fig. 4. The ANOAC receives four sensory inputs from the front (left and right) and side (left and right) of the flying robot to ensure its safety during the exploration task. Three control system outputs from the ANOAC are used to regulate the roll, pitch, and yaw of the robot. This configuration enables the flying robot to perform exploration in complex environments without colliding with obstacles.

3.1. Adaptive neural exploration control (ANEC)

The ANEC consists of the NLCN, post-processing network, and adaptive bias thresholding mechanism (see Fig. 3). These components work together to produce adaptive exploration outputs, as illustrated in Fig. 3(a). The NLCN with two recurrent neurons ($E1$ and $E2$) serves as a neural model³ for generating Lorenz attractor-based chaotic dynamics. An output of the NLCN ($E2$ output) is utilized as a baseline chaotic

³ A detailed explanation of the transformation from the continuous-time Lorenz attractor mathematical model to the discrete-time Lorenz attractor neural model (NLCN) is provided in the Supplementary Material, Section 1.

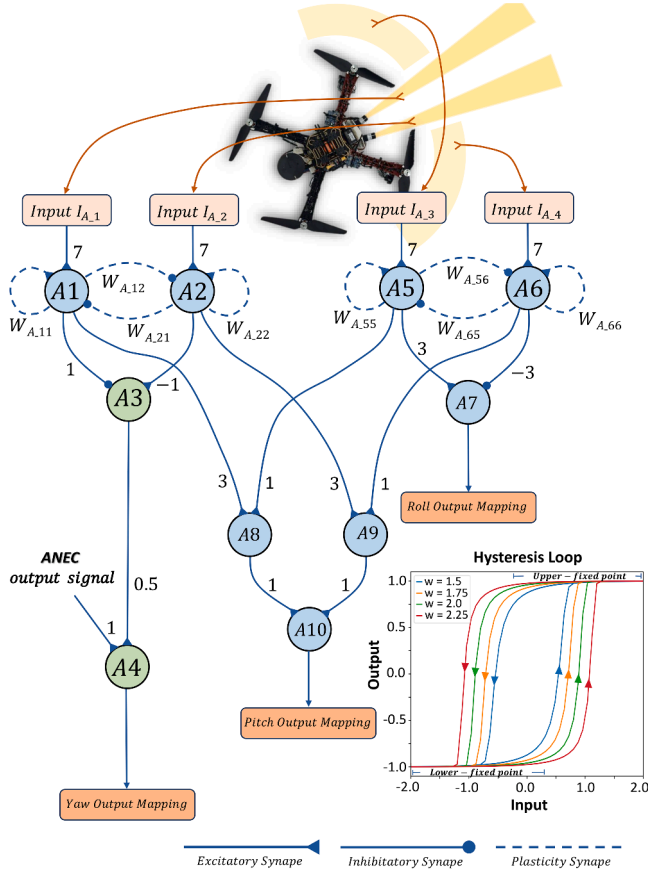


Fig. 4. The ANOAC employs two NHCNs with a synaptic plasticity learning mechanism (illustrated by the dashed lines). The inset figure illustrates the hysteresis (loops) neural dynamics generated by the neurons (A1, A2, A5, or A6) in the NHCN. Here, w_s represents the recurrent weights ($W_{A,11}$, $W_{A,22}$, $W_{A,55}$, or $W_{A,66}$), where the sizes of the hysteresis loops varying according to the values of w_s . By utilizing optimized w_s values, the neurons generate outputs with optimal characteristics by leveraging their hysteresis properties. These outputs are post-processed and used to control the flying robot's roll, pitch, and yaw movements. Note that the input mentioned in the figure refers to the summation of all inputs to the neuron except the recurrent input. For example, the inputs to A1 are calculated as $(I_{A,1} \times 7) + (O_{A2} \times W_{A,21})$, and the output refers to the corresponding neuron output.

signal for exploration behaviors. To translate this chaotic signal into the proper final control command, it is projected to the input neuron E3 of the post-processing network for further processing. This network incorporates an adaptive bias thresholding mechanism in its hidden neurons E4 and E5, which enables the execution of various exploration behaviors. The outputs of E4 and E5 are finally combined in the output neuron E6, and its output is applied to control the flying robot's yaw movement via the ANOAC module. All neurons (E1 – E6 in Fig. 3) employed in this system are modeled as discrete-time, non-spiking neurons. The activity $a_{E,i}$ of each neuron evolves according to the following equation:

$$a_{E,i}(t) = \sum_{j=1}^n W_{E,ij} \cdot o_{E,j}(t-1) + B_{E,i} \quad i = 1, \dots, 6, \quad (1)$$

$$o_{E,i}(t) = a_{E,i}(t), \quad (2)$$

$$o_{E,i}(t) = \begin{cases} a_{E,i}(t) & a_{E,i}(t) > 0, \\ a_{E,i}(t) & a_{E,i}(t) \leq 0, \end{cases} \quad (3)$$

$$o_{E,i}(t) = \tanh(a_{E,i}(t)), \quad (4)$$

where n denotes the number of neurons (here $n = 6$), and $B_{E,i}$ is the internal bias term or sensory input to neuron i . $W_{E,ij}$ represents the synaptic weight of the connection from neuron j to neuron i . The out-

puts $o_{E,j}$ of the neurons are calculated using three different activation functions, which are a linear activation function (Eq. 2) for neurons E1, E2, and E3 (green neurons, Fig. 3(b)), ReLU activation function (Eq. 3) for neurons E4 and E5 (yellow neurons, Fig. 3(c)), and tanh activation function (Eq. 4) for neuron E6 (blue neuron, Fig. 3(c)). Additionally, an overall illustrative diagram of the neuron connections in ANEC, along with the corresponding equations and parameters, is provided in the Supplementary Material, Section 2.

3.1.1. Neuro-Lorenz control network (NLCN)

The NLCN is synthesized as a two-neuron recurrent network, represented by neurons E1 and E2 (Fig. 3(b)). The synaptic weights of the network are derived from the Lorenz system (Li et al., 2016; Lorenz, 1963). The activity of each neuron is described as follows:

$$a_{E1}(t) = W_{E,11} \cdot o_{E1}(t-1) + W_{E,12} \cdot o_{E2}(t-1), \quad (5)$$

$$a_{E2}(t) = W_{E,22} \cdot o_{E2}(t-1) + W_{E,21} \cdot o_{E1}(t-1) + I_a(t-1), \quad (6)$$

where a_{E1} and a_{E2} denote the activities of E1 and E2, respectively. o_{E1} and o_{E2} denote the outputs of E1 and E2, respectively. $W_{E,11}$ represents the recurrent synaptic weight of E1. $W_{E,12}$ represents the synaptic weight of the connection from E1 to E2. $W_{E,22}$ represents the recurrent synaptic weight of E2. $W_{E,21}$ represents the synaptic weight of the connection from E2 to E1. I_a is an adaptive input, which is described by:

$$I_a(t) = -h \cdot o_{E1}(t-1) \cdot S_a(t-1), \quad (7)$$

$$S_a(t) = \psi \cdot S_a(t-1) + h \cdot o_{E1}(t-1) \cdot o_{E2}(t-1), \quad (8)$$

where h denotes a differential step, S_a denotes an internal state of the adaptive input, and ψ is an internal state parameter.

Based on the setup, the neural dynamics of the NLCN resembles the discrete-time Lorenz attractor, as shown in the inset of Fig. 3(a) (see the mathematical formulation translating the Lorenz attractor model to the NLCN in the Supplementary Material, Section 2). This network functions as an intrinsic signal generator for producing complex exploration behaviors. The neural dynamics is illustrated in Fig. 3(a) by plotting the outputs of E1 and E2, along with the internal state of the adaptive input (I_a). In this study, we specifically utilized the network output of E2 as an input for both the post-processing network and adaptive bias thresholding mechanism. Which is detailed in the next section.

3.1.2. Post-processing network and adaptive bias thresholding mechanism

The post-processing network (E3 – E6 in Fig. 3(c)) and adaptive bias thresholding mechanism serve as an adaptive post-processing neural system. The first is neuron E3, which receives inputs from the output of E2 and a bias derived from the average value of the E2 output with an input weight of -1.0. This bias adjusts the zero point, aligning it with the average value of the E2 output. Since both positive and negative values of the input are important to shape the exploration behavior, the output of E3 is divided into two pathways: the positive path and negative path. The positive path transmits the E3 output, with a synaptic weight of 1.0, the input of E4, along with an adaptive bias derived from the standard deviation of the E2 output distribution. The negative path sends the E3 output, with a synaptic weight of -1.0, to the input of E5, also accompanied by the adaptive bias derived from the standard deviation of the E2 output distribution. This adaptive bias acts as a threshold to suppress lower values, facilitated by the ReLU activation function (see Eq. 3) applied at E4 and E5. With this setting, E4 and E5 can be considered positive and negative thresholding neurons, respectively. The outputs of E4 and E5, with synaptic weights of 3.0 and -3.0, respectively, are projected as inputs to E6. These inputs undergo further processing to generate an adaptive exploration output command, where the adaptive bias values and the dynamic behavior of the input signal (i.e., the input to E3) shape the output pattern.

By varying the bias values (1.0σ , 1.5σ , and 2.0σ), different output patterns from E6 are produced, as shown in the blue block of Fig. 3(d).

For instance, applying a bias value of 1.0σ , the lowest used in the experiment, results in an output pattern with the highest number of pulses (both positive and negative) and the longest pulse durations, as illustrated in the upper graph of the blue block in Fig. 3(d). Another important feature is the continuity of the positive or negative pulses, which depends on the dynamic behavior of the input signal received by $E3$ (i.e., the output of $E2$). The output patterns are shaped by three key features that influence the resulting exploration behaviors: (1) the number of pulses, (2) duration (or width) of each pulse, and (3) continuity of positive or negative pulse sequences. These three features depend on two factors, which are the bias value and dynamic behavior of the input signal received by $E3$. The output of $E6$ is then used to control the flying behavior of the autonomous flying robot, in conjunction with the output from the ANOAC module (described in the next section). In this study, we defined that a positive output value drives the robot to rotate clockwise (turning right) along its own axis, while a negative value causes it to rotate counterclockwise (turning left). Consequently, the robot's flight pattern dynamically adapts in response to variations in the $E6$ output. In summary, based on the three key features, the number of pulses determines the frequency of the robot's turns: more pulses lead to more frequent turns, and fewer pulses to less frequent turns. The duration (or width) of each pulse controls the turning angle: a larger width results in a larger angle, and a smaller width in a smaller angle. Furthermore, the continuity of positive or negative pulse sequences dictates the number of consecutive turns in the same direction, producing a cumulative turning effect and larger net angle. Through this adaptive control mechanism, the system generates complex exploratory flight behaviors that effectively combine both local and global search strategies, leveraging the NLCN and its adaptive bias thresholding mechanism.

Furthermore the generated exploration behavior in the flying robot, specifically the step length (or flying distance), exhibits a pattern that mirrors the step length (l_s) seen in the Lévy flight search model with respect to changes in the output of $E6$. To substantiate this similarity between the step lengths, we performed an analysis based on the power-law (Lévy) distribution. For simplicity, the analysis is conducted in two stages. The first stage involves estimating the power-law exponent (α) as described in Aaron et al. (2009). The second stage calculates the goodness-of-fit between the empirical data from the ANEC and power-law (Lévy) model using the Kolmogorov-Smirnov (KS) test. In this context, the p-value represents the probability of obtaining a KS statistic at least as large as the one observed, under the null hypothesis that both distributions are the same. This approach allows for a comparison between the intrinsic NLCN-based ANEC system and mathematical power-law (Lévy) model (refer to the Experiments and Results section for details).

3.2. Adaptive neural obstacle avoidance control (ANOAC)

The ANOAC proposed in this study is an extension of the system introduced in a previous study (Pedersen & Manoonpong, 2018), which was designed solely for forward obstacle detection. In contrast, the present work extends this approach to handle both forward and lateral (sideward) obstacle detection, thus offering a more comprehensive, all-around comprehensive ANOAC (as illustrated in Fig. 4). This expanded ANOAC is integrated with the ANEC system to enable the flying robot to perform safe exploration in complex environments. The proposed ANOAC consists of two key components: the neuro-hysteresis control network (NHCN) and synaptic plasticity mechanism. All neurons in this network are modeled as discrete-time, non-spiking neurons, and they utilize the hyperbolic tangent (tanh) activation function as described in Eqs. (1) and (4), except for neuron A4, which employs a linear activation function as specified in Eq. 2. Additionally, an overall illustrative diagram of the neuron connections in ANOAC, along with the corresponding equations and parameters, is provided in the Supplementary Material, Section 3.

3.2.1. Neuro-hysteresis control network (NHCN)

The NHCN is designed as a two-neuron recurrent network, producing outputs that exhibit neural dynamics with a hysteresis effect. The magnitude of the hysteresis loop (illustrated in the inset of Fig. 4) depends on the recurrent synaptic weight values within the network. This hysteresis effect functions as an adaptive short-term memory mechanism, enabling the system to prolong the turning movement to avoid obstacles across different environmental settings, such as areas with low or high obstacle density or regions with deadlocks. In this study, two NHCNs were employed: one generates a steering (yaw) control command (A1 and A2 in Fig. 4), based on two forward obstacle detection sensors, while the other produced a steering (roll) control command (A5 and A6 in Fig. 4), based on two sideward obstacle detection sensors. The neurons' activities in the NHCN responsible for generating the yaw control command are described below:

$$a_{A1}(t) = W_{A,11} \cdot o_{A1}(t-1) + W_{A,12} \cdot o_{A2}(t-1) + W_{I_1} \cdot I_{A,1}(t), \quad (9)$$

$$a_{A2}(t) = W_{A,22} \cdot o_{A2}(t-1) + W_{A,21} \cdot o_{A1}(t-1) + W_{I_2} \cdot I_{A,2}(t), \quad (10)$$

where a_{A1} and a_{A2} denote the activities of A1 and A2, respectively; o_{A1} and o_{A2} denote the outputs of A1 and A2, respectively; $W_{A,11}$ represents the recurrent synaptic weight of A1; $W_{A,12}$ represents the synaptic weight of the connection from A1 to A2; $W_{A,22}$ represents the recurrent synaptic weight of A2; $W_{A,21}$ represents the synaptic weight of the connection from A2 to A1; $I_{A,1}$ is an input from the left sensor; $W_{I_{A,2}}$ is the synaptic weight of input $I_{A,1}$; $I_{A,2}$ is an input from the right sensor; and $W_{I_{A,2}}$ is the synaptic weight of input $I_{A,2}$.

The neurons' activities in the NHCN generating the roll control command are described below:

$$a_{A5}(t) = W_{A,55} \cdot o_{A5}(t-1) + W_{A,56} \cdot o_{A6}(t-1) + W_{I_{A,3}} \cdot I_{A,3}(t), \quad (11)$$

$$a_{A6}(t) = W_{A,66} \cdot o_{A6}(t-1) + W_{A,65} \cdot o_{A5}(t-1) + W_{I_{A,4}} \cdot I_{A,4}(t), \quad (12)$$

where a_{A5} and a_{A6} denote the activities of A5 and A6, respectively; o_{A5} and o_{A6} denote the outputs of A5 and A6, respectively; $W_{A,55}$ represents the recurrent synaptic weight of A5; $W_{A,56}$ represents the synaptic weight of the connection from A5 to A6; $W_{A,66}$ represents the recurrent synaptic weight of A6; $W_{A,65}$ represents the synaptic weight of the connection from A6 to A5; $I_{A,3}$ is an input from the lateral right sensor; $W_{I_{A,3}}$ is the synaptic weight of input $I_{A,3}$; $I_{A,4}$ is an input from the lateral left sensor; and $W_{I_{A,4}}$ is the synaptic weight of input $I_{A,4}$.

The core of this system is the NHCNs with synaptic plasticity mechanism. The synaptic weights ($W_{A,11,A,22}$, $W_{A,12,A,21}$, $W_{A,55,A,66}$, and $W_{A,56,A,65}$) are plasticity developed based on Hebbian correlation-based learning, explained in the next section. Figure 4 shows the neuro-computational process in which obstacle detection and avoidance are integrated with the exploration behavior of the flying robot. The process begins with sensors detecting obstacles and transmitting the input signals to the NHCNs. Two input signals from the front sensors (ranging from $[-1, 1]$) are sent to A1 and A2. These signals are processed through A3, and the resulting output is integrated with the output from the ANEC at A4 (refer to Section 3.3 for integration details). The output from A4 controls the yaw of the flying robot with a positive value indicating a clockwise turn and negative value indicating a counterclockwise turn. Simultaneously, the two input signals from the lateral sensors (mapped to the range $[-1, 1]$) are transmitted to neurons A5 and A6 for processing. The outputs of A5 and A6 are combined at A7, generating the roll control signal. A positive value causes the robot to shift right, whereas a negative value causes it to shift left. To process a pitch control command, the outputs from A1 and A5 are transmitted to A8 for post-processing, while the outputs from A2 and A6 are transmitted to A9. The outputs from both A8 and A9 are then combined at A10 to produce the final pitch control, where a positive value corresponds to backward movement and negative value to forward movement.

3.2.2. Synaptic plasticity mechanism

The synaptic plasticity mechanism (Grinke et al., 2015a) is based on online Hebbian correlation-based learning. The learning rule is realized

by three factors: the output activity $o_{Ai}(t)$ of neurons (specifically, A1, A2, A5, and A6) at time step t , output activity $o_{Ai}(t-1)$ of the same neurons at the previous time step ($t-1$), and reflex signal $R_{A,i}(t)$. The reflex signal, utilized to control the learning process, is computed from the sensory input $I_{A,i}$ as below:

$$R_{A,i}(t) = \begin{cases} 1 & I_{A,i}(t) > -0.5, \quad i = 1, \dots, 4, \\ 0 & I_{A,i}(t) \leq -0.5, \quad i = 1, \dots, 4, \end{cases} \quad (13)$$

where $R_{A,i}(t)$ is the reflex signal and $I_{A,i}(t)$ is the input signal of sensor i .

To ensure that the synaptic weights maintain their stability and prevent sign changes, which could disrupt the hysteresis effects, we implemented a stability constraint based on prior studies (Tetzlaff et al., 2011, 2012). We mapped the outputs o_{Ai} which are in the range of $[-1, 1]$ to the positive interval v_{Ai} $[0, 1]$ as below:

$$v_{Ai}(t) = 0.5 \cdot o_{Ai}(t) + 0.5, \quad (14)$$

where $v_{Ai}(t)$ is the positive mapped output of $o_{Ai}(t)$ and i refers to 1, 2, 5, and 6.

The recurrent excitatory weights ($W_{A,11,A,22}$ and $W_{A,55,A,66}$) are governed as follows:

$$\Delta W_{A,ii} = \mu \cdot v_{Ai}(t-1) \cdot v_{Ai}(t) \cdot R_{A,i}(t) + \gamma \cdot (k - v_{Ai}(t)) \cdot W_{A,ii}(t)^2, \quad (15)$$

$$W_{A,ii}(t+1) = W_{A,ii}(t) + \Delta W_{A,ii}, \quad (16)$$

where $\Delta W_{A,ii}$ is the weight value change in a time step; μ is a learning rate for the excitatory synapses, set to 0.01; γ is a forgetting rate, set to 0.0003; k is an offset that enables a constant reduction of the weights when no obstacle is detected, set to -0.01; and i represents 1, 2, 5, and 6.

The inhibitory weights ($W_{A,12,A,21}$ and $W_{A,56,A,65}$) were developed in a similar manner but were changed equally to maintain symmetry. This was achieved by using the average changed value of $W_{A,12,A,21}$ to update both weight values of $W_{A,12,A,21}$ and applying a similar process to $W_{A,56,A,65}$, as follows:

$$\Delta W_{T,ij} = \mu \cdot v_{Ai}(t-1) \cdot v_{Aj}(t) \cdot R_{A,i}(t) + \gamma \cdot (k - v_{Ai}(t)) \cdot W_{T,ij}(t)^2, \quad (17)$$

$$\Delta W_{T,ji} = \mu \cdot v_{Aj}(t-1) \cdot v_{Ai}(t) \cdot R_{A,j}(t) + \gamma \cdot (k - v_{Aj}(t)) \cdot W_{T,ji}(t)^2, \quad (18)$$

$$W_{T,ij}(t+1) = W_{T,ij}(t) - \Delta W_{T,ij}, \quad (19)$$

$$W_{T,ji}(t+1) = W_{T,ji}(t) - \Delta W_{T,ji}, \quad (20)$$

$$W_{A,ij}(t+1), W_{A,ji}(t+1) = 0.5W_{T,ij}(t) + 0.5W_{T,ji}(t), \quad (21)$$

where $\Delta W_{T,ij}$ and $\Delta W_{T,ji}$ are temporary variables for calculating the average changed values of the inhibitory weights; $W_{T,ij}$ and $W_{T,ji}$ are temporary variables for calculating the average inhibitory weights; μ is a learning rate for the inhibitory synapses, set to 0.015; γ is a forgetting rate and k is an offset, both are set to the same values as above; and i represents 1 and 5, while j represents 2 and 6.

Fundamentally, this learning mechanism develops the synaptic weights of the two-neuron recurrent networks in two key aspects. Firstly, when the flying robot is trying to avoid an obstacle or escape from a corner or a deadlock, the sensory inputs and recurrent excitatory weights ($W_{A,11,A,22}$ and $W_{A,55,A,66}$) work together to drive the neural output to reach and stay at the upper fixed point (approximately +1) of the hysteresis loop (see Fig. 4). Simultaneously, as the output becomes active, the activation of the nearby neuron is temporarily inhibited due to the effect of inhibitory weights ($W_{A,12,A,21}$ and $W_{A,56,A,65}$) connecting from the output to the input of the nearby neuron. In this state, when the flying robot gets stuck in a situation, all weights are in the learning process. Both the excitatory weights ($W_{A,11,A,22,A,55,A,66}$) and inhibitory weights ($W_{A,12,A,21,A,56,A,65}$) will learn and grow (based on their sensory inputs) until they achieve appropriate values. Consequently, the flying robot successfully executes a significantly large turning angle to escape from the situation. For instance, if the robot gets stuck in a corner, the front left sensor (input $I_{A,1}$) detects an obstacle and the A1 output becomes active, resulting in temporarily inhibiting the A2 output. With the hysteresis properties, the A1 output will remain active for a longer period (depending on the weight values),

causing the flying robot to perform a larger turning angle and navigate away from the corner. Secondly, when the flying robot has successfully escaped or no longer detects obstacles, the neural output returns to the lower fixed point (approximately -1) of the hysteresis loop (see Fig. 4). Consequently, the robot stops turning and resumes forward flight. Simultaneously, the forgetting process begins, during which the synaptic weight values gradually decrease until the next obstacle is detected or the robot gets stuck again. This triggers the activation of the reflex signal. The weight values will then be significantly readjusted, demonstrating the adaptive nature of the system. Thus, the learning dynamics driven by the synaptic plasticity mechanism depends on the robot-environment interaction realized through sensory feedback. This enables the robot to avoid obstacles efficiently; for example, making small turns for small obstacles or significantly larger turns with hysteresis-driven action prolongation for larger or deadlock-shaped obstacles. Thus, the robot can explore safely and efficiently without getting stuck in challenging environments. From a learning perspective, the synaptic plasticity mechanism allows the system to continuously adjust synaptic weights in real-time based on sensory feedback, balancing learning and forgetting to gradually converge toward optimal values for effective obstacle avoidance. This control mechanism also provides a possible solution for online or lifelong robot learning in real-world applications. Additionally, the detailed analysis of the effect of learning and forgetting rates can be found in the Supplementary Material, Section 4.

3.3. Integrated adaptive neural control system (IANCS)

To achieve safe adaptive exploration behavior with safety in the flying robot, two modular neural control systems, each with distinct neural dynamics, were integrated (see Fig. 2 and Algorithm 1). The first system is the ANEC, which consists of the NLCN coupled with the adaptive bias thresholding mechanism. The ANEC serves as the internal driving mechanism for generating adaptive exploratory signals. These signals are processed and subsequently utilized to control the robot's left and right turns, enabling the execution of both local and global search and exploration tasks within a defined area of interest. The adaptability of exploration behavior is facilitated by the modulation of the bias value via the adaptive bias thresholding mechanism, allowing the flying robot to execute a range of various exploration strategies. The second system is the ANOAC, which integrates two NHCNs and the synaptic plasticity mechanism. The ANOAC is responsible for ensuring the safe execution of search and exploration tasks, by avoiding obstacles. The ANOAC is capable of learning and dynamically adapting obstacle avoidance behavior online, depending on the environmental conditions such as obstacle density, corners, and deadlocks.

To ensure that both exploration and obstacle avoidance behaviors work seamlessly in tandem, the two modules (ANEC and ANOAC) are integrated using the algorithm described in Algorithm 1. This algorithm governs the robot's decision-making process by controlling the roll, pitch, and yaw movements based on real-time sensory inputs and outputs of the neural control systems. To integrate the neural control systems (ANEC and ANOAC), where each exhibits distinct inherent neural dynamics, the update frequencies of the two systems are empirically set at different rates. The ANOAC, characterized by slower neural dynamics, operates at a standard frequency of 20 Hz. In contrast, the ANEC, which exhibits faster neural dynamics, operates at a 50 times slower rate. This different configuration ensures that the flying robot can effectively respond to both obstacle avoidance and exploration commands, optimizing the system's overall efficiency and responsiveness to environmental changes.

4. Experiments and results

To assess and evaluate the adaptability and performance of the proposed IANCS, three experiments were conducted. The first experiment

Algorithm 1 IANCS.

```

1: Initialize all variables           ▷ see Supplementary Tables 1 and 2
2:  $E1, E2, S_a \sim \mathcal{U}(-1.0, 1.0), h = 0.01, \psi = 0.973$ 
3:  $W_{E,11} = 0.90, W_{E,22} = 0.99, W_{E,12} = 0.28, W_{E,21} = 0.10$ 
4:  $W_{A,11}, W_{A,22}, W_{A,55}, W_{A,66} = 0.5$ 
5:  $W_{A,12}, W_{A,21}, W_{A,56}, W_{A,65} = -0.5$ 
6: while Terminate is not TRUE do
7:   Get inputs  $I_{A,1}, I_{A,2}, I_{A,3},$  and  $I_{A,4}$  ▷ receive input information
8:   Update neurons  $E1$  and  $E2$            ▷ generate neuro-Lorenz signal
9:   Update  $I_a$  and  $S_a$                  ▷ compute adaptive input and internal state
10:  Update  $\bar{X}$  and  $\sigma$                  ▷ compute distribution parameters
11:  Update neurons  $A1, A2, A5,$  and  $A6$    ▷ compute obstacle avoidance
12:  Update neurons  $A3, A7, A8, A9,$  and  $A10$  ▷ post processing
13:  Update neuron  $A4 \leftarrow 0.5A3 + E6$  ▷ Update final yaw command
14:  Roll Command  $\leftarrow A7$  output       ▷ Roll control output
15:  Pitch Command  $\leftarrow A10$  output     ▷ Pitch control output
16:  Yaw Command  $\leftarrow A4$  output       ▷ Yaw control output
17:  Update weights  $W_{A,11}, W_{A,22}, W_{A,12},$  and  $W_{A,21}$  ▷ synaptic plasticity
18:  Update weights  $W_{A,55}, W_{A,66}, W_{A,56},$  and  $W_{A,65}$  ▷ synaptic plasticity
19:  if Count < 50 then
20:    Count ++
21:  else
22:    Update neurons  $E3, E4, E5,$  and  $E6$  ▷ compute adaptive thresholding output (based on neuro-Lorenz signal)
23:    Count = 0
24:  end if
25:  if Global search is TRUE then           ▷ perform global search
26:    if Search_count < 15000 then         ▷ local search condition
27:      bias  $\leftarrow 1.0\sigma$              ▷ LOW bias
28:      Search_count ++
29:    else if Search_count < 30000 then ▷ global search condition
30:      bias  $\leftarrow 2.0\sigma$              ▷ HIGH bias
31:      Search_count ++
32:    else
33:      Search_count = 0
34:    end if
35:  else                                     ▷ perform local search
36:    bias  $\leftarrow 1.0\sigma$              ▷ LOW bias
37:  end if
38: end while

```

focused on demonstrating the system's adaptability during exploration missions in an open environment. This experiment involved analysis and comparison with the mathematical Lévy flight (power-law) model to evaluate the system's exploratory behavior. The second experiment aimed to showcase the flying robot's behavior in a closed environment with various obstacle configurations. Additionally, it investigated the impact of the exploration performance and learning process of the synaptic plasticity mechanism on the ANOAC. Finally, the practical utility of the system was demonstrated through the use of a real flying robot, showcasing its capabilities in real-world scenarios.

4.1. Experimental setup

The integrated neural control system was implemented on both simulated and real flying robot platforms. For the simulated environment, the CoppeliaSim platform was used with several distinct environmental settings, as depicted in Fig. 5(a)–(f). The settings were as follows:

- (a) the simulated flying robot in the CoppeliaSim platform,
- (b) an open obstacle-free environment (open space),
- (c) 10x10 m² closed environment with obstacle-free,

- (d) 10x10 m² closed environment with low obstacle density,
- (e) 10x10 m² closed environment with high obstacle density,
- (f) 10x10 m² closed environment with deadlock-shaped obstacles.

The simulated flying robot (shown in Fig. 5(a)) used in these experiments was equipped with two frontal infrared sensors, each having a detection range of 0.75 m, and two side-surrounding infrared sensors, each with a detection range of 0.15 m (measured from the edge of the robot). These four sensors were used as inputs for the ANOAC to ensure the flying robot's safety by preventing collisions during exploration missions. The Robot Operating System (ROS) was employed to facilitate communication between the flying robot and neural control system.

For the real flying robot, we utilized an F450 frame wheel quadrotor as the experimental platform. The robot was equipped with four MN2212-21 kV brushless motors, each paired with a 20 A electronic speed controller. Figure 5(g) illustrates the flying robot's top view and sensor detection views, with yellow representing the LiDAR sensors (detection range 7 m) and orange representing the ultrasonic sensors (detection range 2 m with a 75-degree cone shape). Figure 5(h) provides a detailed view of the real flying robot's components:

- (1) flying robot hardware as described earlier,
- (2) pixhawk Cube Black flight controller (open-source) with its accessories, including GPS for outdoor positioning, telemetry for communication between the flying robot and ground control station, a receiver for manual control signals from the remote controller, and remote controller itself for manual operation,
- (3) an on-board computer (NanoPC T4) for processing the neural control system and facilitating communication with the Pixhawk and sensor systems through an Arduino Nano,
- (4) the Arduino Nano microcontroller, serving as the interface hardware between the sensors and on-board computer,
- (5) two compact optical distance measurement sensors (LiDAR-Lite V3) for detecting frontal obstacles,
- (6) two ultrasonic distance sensors (JSN-SR04T) for detecting side-ward obstacles.

The real flying robot was tested in an outdoor environment size of approximately 12,000 m² (shown in Fig. 5(i) and (j)) using ROS to manage communication between the flying robot and neural control system.

4.2. Adaptive exploration behavior

The first experiment demonstrated the system's adaptability in terms of the flying trajectory of the robot using the proposed IANCS as a navigational controller for exploration in an open environment. To generate adaptive exploration behavior, the system initiated with the NLCN (comprising neurons $E1$ and $E2$), which produced output signals exhibiting dynamics similar to the Lorenz attractor. The output signal from $E2$ was then directed into the post-processing network and the adaptive bias thresholding mechanism. During this process, the bias values (at neurons $E4$ and $E5$ in the post-processing network) were adjusted by alternating between 1.0σ and 2.0σ . Consequently, the post-processed output patterns were adapted based on the changes in bias values. This output was subsequently applied to control the flying robot's movements to explore the target environment. Figure 6(a) shows the tracked trajectory of the flying robot during its exploration of the open area. The trajectory illustrates that variable bias values induce distinct exploration behaviors in the flying robot. With a low bias value (1.0σ), the flying robot frequently executes turns, resulting in a circular flight pattern that focuses on specific regions of the area, representing a behavior akin to a local search. In contrast, when using high bias values (2.0σ), the robot turns less frequently, covering more ground as it flies forward. This behavior represents a global search strategy, where the robot explores a larger area. Therefore, by adapting the bias values to both small and large values, the system is capable of automatically generating adaptive exploration behaviors that combine local and global search strategies.

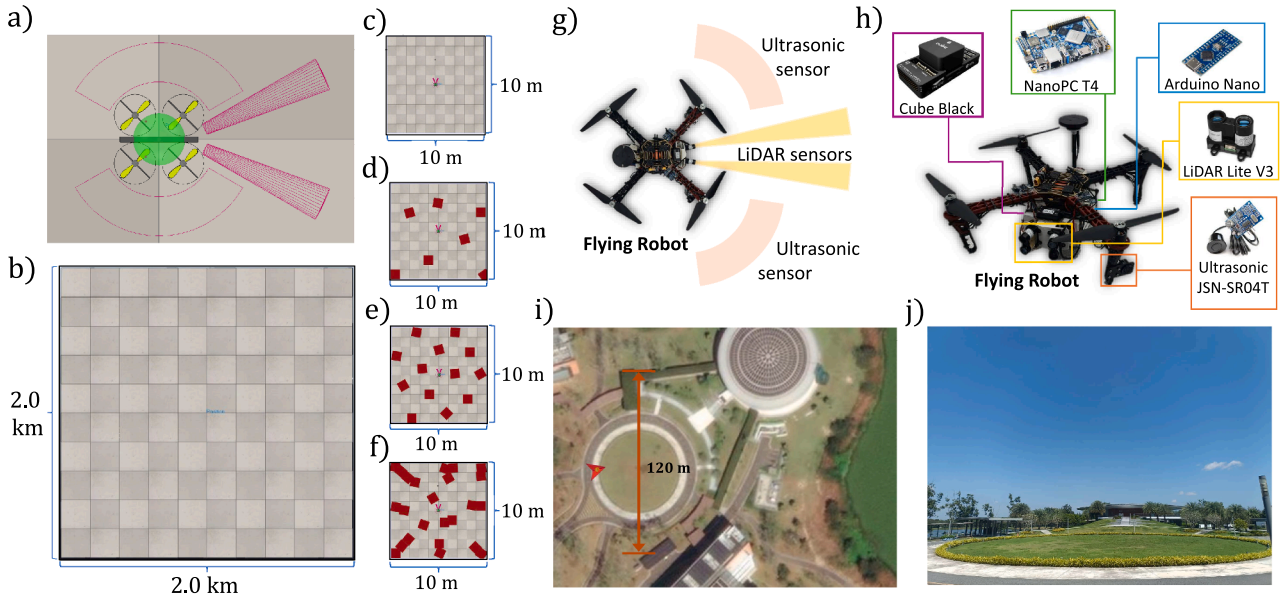


Fig. 5. (a) The simulated flying robot in the CoppeliaSim platform, (b) an obstacle-free open environment (a 2.0x2.0 km² open space), (c) a 10x10 m² obstacle-free closed environment, (d) a 10x10 m² closed environment with a low-density of obstacles, (e) a 10x10 m² closed environment with a high-density of obstacles, (f) a 10x10 m² closed environment with deadlock-shaped obstacles, (g) a real flying robot setting showing the flying robot's top view and sensors' detection view (yellow for LiDAR sensors and orange for ultrasonic sensors), (h) real flying robot components, and (i) an outdoor experimental area in top view and (j) side view.

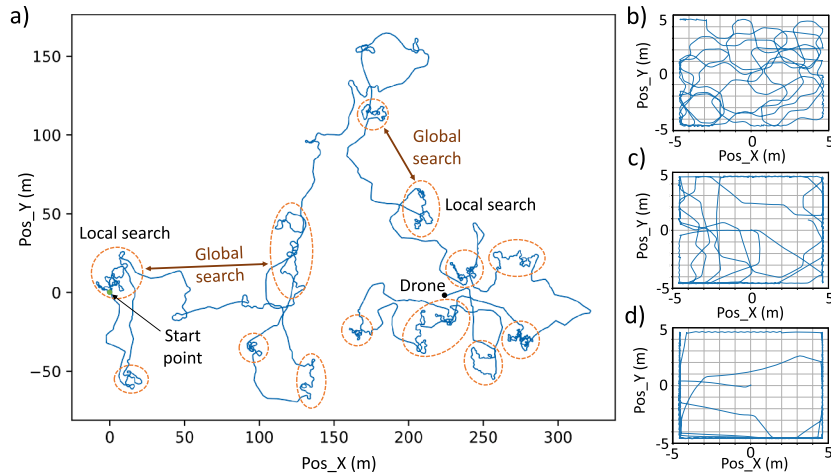


Fig. 6. (a) The flying robot's tracked position while performing adaptive exploration behavior, encompassing both local and global searches, exhibits patterns reminiscent of a Lévy flight. Comparison of the average flying distance while flying the robot in a closed environment, using the IANCS with three different bias values: (b) 1.0 σ , (c) 1.5 σ , and (d) 2.0 σ .

Additionally, the exploration behavior of the flying robot in a closed environment measuring 100 m² (10 m $\times$ 10 m) was assessed using three different bias values (1.0 σ , 1.5 σ , and 2.0 σ), as shown in Fig. 6. The results demonstrate that varying the bias values induced different exploration behaviors. A low bias value (1.0 σ) led to the flying robot making frequent turns, resulting in a circular flight pattern that focused on specific regions of the environment (Fig. 6(b)). In contrast, higher bias values (1.5 σ and 2.0 σ) caused the robot to execute fewer turns and exhibit wall-following behavior, which is evident in the enclosed spaces (Fig. 6(c) and (d)). Furthermore, to investigate the similarity between the Lévy flight (power-law) model and empirical data generated by ANEC from the experiment conducted in the open environment (Fig. 6(a)), we performed a statistical analysis using the Kolmogorov-Smirnov (KS) test. The results revealed that there was no significant difference ($p > 0.05$) between the distributions of the Lévy flight (power-law with $\alpha = 1.5$) model and empirical data generated by the ANEC.

Figure 7(a) shows the distribution histogram of the Lévy flight (power-law with $\alpha = 1.5$), while Fig. 7(b) displays the distribution histogram of the ANEC-generated exploration data. As seen in Fig. 7, the ANEC played a key role in driving the flying robot's exploration behavior, as demonstrated in Fig. 6. Finally, Fig. 7(c) presents a comparison of the cumulative distribution functions (CDFs) between the Lévy flight (power-law with $\alpha = 1.5$) model and ANEC on a logarithmic scale for a more direct comparison of the distributions.

4.3. Adaptive obstacle avoidance behavior

The second experiment focused on exploring the system's adaptability and performance in obstacle avoidance by integrating two adaptive neural control systems: the ANOAC and ANEC (IANCS). In this experiment, the flying robot was tested in a designated area of 100 m² (10 m $\times$ 10 m), simulating three different closed environments: low obstacle

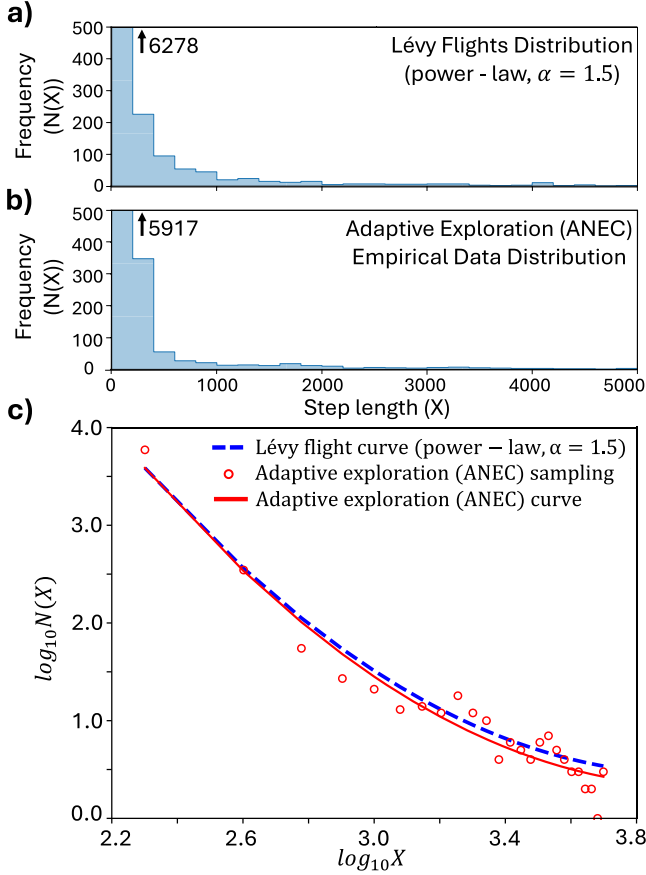


Fig. 7. (a) Distribution histogram of the Lévy flight (power-law with $\alpha = 1.5$) model. (b) Distribution histogram of the ANEC, which played a central role in driving the flying robot's exploration behavior. (c) Comparison of the cumulative distribution functions (CDFs) between the Lévy flight (power-law with $\alpha = 1.5$) model and the ANEC in a logarithmic scale shows no significant difference between the two distributions ($p > 0.05$).

density (LOW), high obstacle density (HIGH), and very high obstacle density with deadlocks (DEADLOCK). For each environment, the robot flew for 30 minutes from randomly assigned initial positions and heading directions, while maintaining a fixed initial velocity. Each experiment was repeated ten times for each environment to ensure reliability. Notably, for this experiment, the bias value applied in the adaptive bias thresholding mechanism of the ANEC was 1.0σ , focusing primarily on local exploration performance. The results in Fig. 8 illustrate the flying behavior of the robot and development of synaptic weights in the ANOAC's synaptic plasticity mechanism while navigating three distinct closed environments: LOW, HIGH, and DEADLOCK using the integration of ANOAC and ANEC (IANCS). Additionally, to assess the impact of varying initial flying speeds, separate experiments were conducted under different speed settings. The corresponding results are provided in the Supplementary Material, Section 5.

Figure 9(a) shows the ablation test results comparing the average flying distances achieved by three exploration control systems: ANEC, which includes only the exploration control module without obstacle collision protection; ANOAC, which consists solely of the adaptive obstacle avoidance module and issues a fly-forward exploration command when no obstacle is detected; and IANCS (integration of ANOAC and ANEC). The results show that the control system using only ANEC achieves the lowest average flying distances – 7.5 m, 7.2 m, and 4.8 m in the LOW, HIGH, and DEADLOCK environments, respectively – as the robot frequently collides with obstacles and falls due to the absence of any obstacle avoidance capability. In contrast, the system using only ANOAC demonstrates improved performance, achieving average distances of 70.5 m, 55.2 m, and 67.3 m in the LOW, HIGH, and DEADLOCK environments, respectively. This improvement is attributed to the robot's ability to avoid collisions and continue exploring for longer periods. However, the robot often becomes trapped in corners or deadlock situations, requiring significant synaptic weight adaptation to escape. During these situations, the robot tends to oscillate in place while the synaptic weights gradually evolve toward values that enable escape (see Fig. 9(a) and (b)). Notably, the integration of ANOAC and ANEC yielded the best performance, with average distances of 119.5 m, 105.7 m, and 103.0 m in the LOW, HIGH, and DEADLOCK environments, respectively. This is because the collaboration between the two neural control mechanisms allows the robot to escape difficult situations more efficiently.

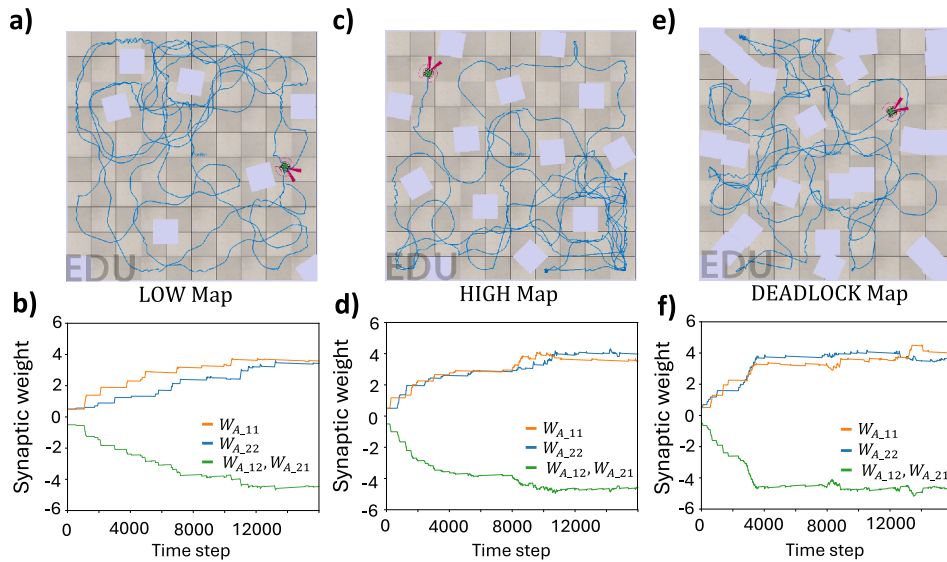


Fig. 8. Snapshot flying behavior of the flying robot while flying in a closed environment and weight development of the synaptic plasticity mechanism in the ANOAC. (a) and (b) Represent the flying behavior and weights in a low obstacle density (LOW) environment. (c) and (d) Represent the flying behavior and weights in a high obstacle density (HIGH) environment. (e) and (f) Represent the flying behavior and weights in a very high obstacle density and deadlocks (DEADLOCK) environment.

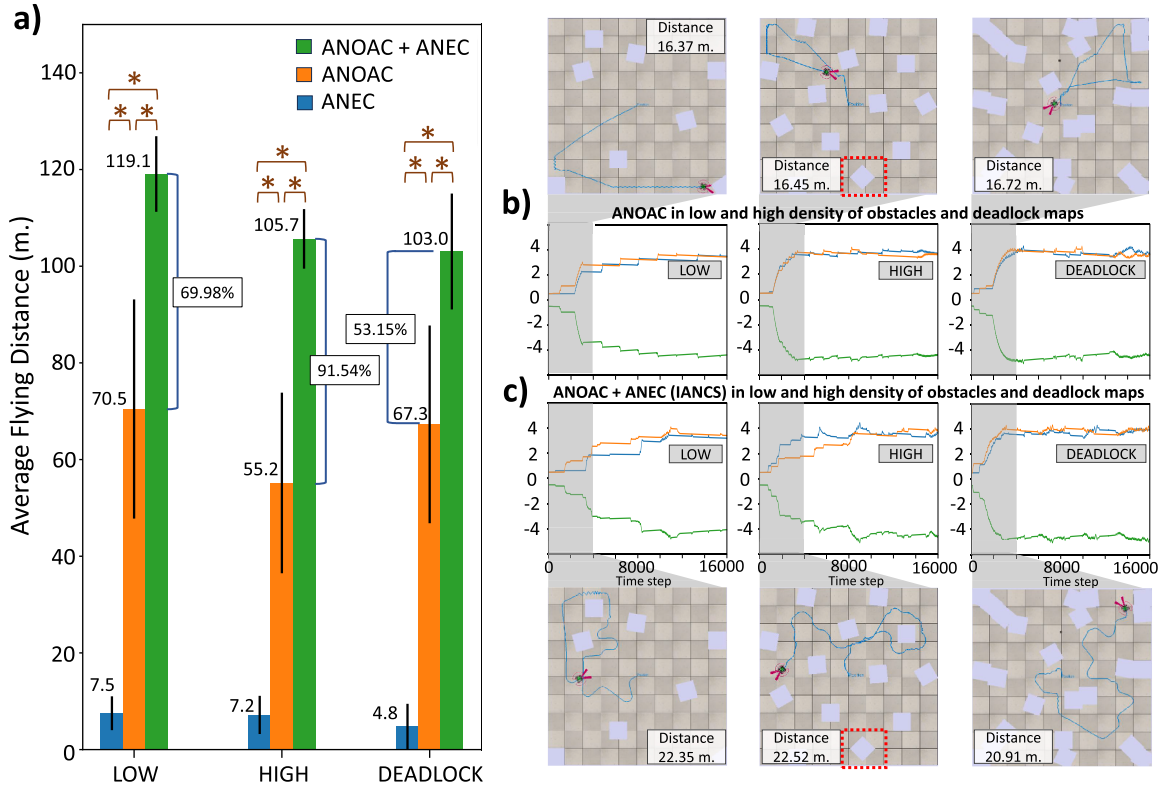


Fig. 9. (a) Comparison of the average flying distance from the ablation test across the three environments, LOW, HIGH, and DEADLOCK, using three exploration control systems: ANEC, ANOAC, and IANCS (ANOAC + ANEC). * Denotes a significant differences with $p \leq 0.05$. (b) and (c) Show comparisons of synaptic weight development and flying trajectory snapshots during flights in three distinct closed environments: low obstacle density (LOW), high obstacle density (HIGH), and very high obstacle density with deadlocks (DEADLOCK). The evaluation was conducted using exploration control systems based on ANOAC alone (b) and the IANCS (c). In the graphs, the synaptic weights of ANOAC are represented by the following colors: orange for $W_{A,11}$, blue for $W_{A,22}$, and green for $W_{A,12,A,21}$. Shaded gray areas indicate periods of weight development, corresponding to the flying behavior shown in the snapshot figures.

While ANOAC adapts the synaptic weights, ANEC simultaneously generates turning commands that enable the robot to maneuver out of corners and deadlocks more effectively. To further quantify this improvement, ANOAC alone was used as a baseline for comparison. The combined system (ANOAC + ANEC) achieved average flying distance increases of 69.98%, 91.54%, and 53.15% in the LOW, HIGH, and DEADLOCK environments, respectively, as shown in Fig. 9(a). Interestingly, the average flying distance was the lowest for the robot using only the ANOAC (orange) in the high obstacle density (HIGH) environment, as it spent a significant amount of time stuck in a corner with a large discontinuous edge between an obstacle and wall (red dashed block in Fig. 9(b) and (c)). This discontinuous edge was large enough to eliminate the hysteresis effect and disrupt the learning process, but not large enough for the robot to fly through. Consequently, when the robot encountered this situation, it required additional time to adapt the synaptic weights compared to situations with continuous edges or smaller discontinuous edges, leaving less time for exploration. With the integrated control system (IANCS), however, the robot achieved the highest percentage improvement in this environment, increasing its average distance to 105.7 m (+ 91.54%). Moreover, the proposed integration accelerates synaptic weight convergence toward optimal values, enabling more stable and effective navigation. Further details and an in-depth analysis of this enhanced learning process are provided in the Supplementary Material, Section 6.

4.4. State-of-the-art comparison

Next, we compared the proposed integrated adaptive neural control system (IANCS), which combines the ANEC with the ANOAC, with several state-of-the-art exploration strategies. These include bio-inspired

approaches such as Lévy flight and random walk, as well as a data-driven learning-based method (i.e., deep reinforcement learning (DRL)). The evaluation considered two key aspects: local exploration performance and computational efficiency. Local search performance was assessed based on the total flying distance achieved while exploring different closed environments with varying obstacle configurations, within a fixed time limit of 30 minutes. The robot operated in a 100 m² (10 m × 10 m) area under three simulated conditions: LOW, HIGH, and DEADLOCK, with randomized initial positions and heading directions. Computational efficiency was evaluated by measuring the computation time per iteration and analyzing the algorithmic time complexity (Big O notation) during both the learning and deployment phases. The results presented in Table 1(a) indicate that the proposed method achieved the highest performance across all environments compared to the Lévy flight and random walk exploration strategies, except in the LOW environment, where the performance of the random walk method was not significantly different. Additional comparison bar graphs and detailed explanations are provided in the Supplementary Material, Section 7. In terms of computational efficiency in Table 1(b), all three methods demonstrated low processing times, approximately 0.1 - 0.2 ms per iteration, with a time complexity of $O(N)$ during both the learning and deployment phases. In contrast, DRL required significantly more time, taking approximately 44.78 ms during learning and 18.13 ms during deployment, with a time complexity of $O(E \cdot U \cdot N \cdot H^2)$ for learning and $O(N \cdot H^2)$ for deployment, where E is the number of training epochs, U the number of PPO update epochs, N the number of steps per training epoch (i.e., trajectory length), and H the hidden layer size in the neural networks. For more details, see Sections 8 and 9 of the Supplementary Material.

Table 1

The performance comparison table presents results from two perspectives: (a) Local exploration performance, measured by the flying distances achieved using five different exploration strategies—the proposed IANCS method (ANEC with ANOAC), Lévy flight, random walk, and deep reinforcement learning (DRL). * Indicates a statistically significant difference at $p \leq 0.05$, compared to the proposed method (ANEC with ANOAC). (b) Computational efficiency, which includes the computational time per iteration and the overall time complexity (Big O notation) during both the learning and deployment processes. Note that the proposed method was implemented using online learning in real-time deployment.

	Exploration methods			
	IANCS (ANEC + ANOAC)	Lévy flight	Random walk	DRL
a) Local Exploration Performance				
LOW	119.1 ± 7.82 m	89.2 ± 35.52 m*	112.3 ± 11.07 m	105.04 ± 13.02 m*
(% different)	(baseline)	(-25.11%)*	(-5.71%)	(-11.50%)*
HIGH	105.7 ± 6.16 m	54.7 ± 32.26 m*	75.4 ± 17.15 m*	99.17 ± 7.68 m
(% different)	(baseline)	(-48.25%)*	(-28.67%)*	(-6.18%)
DEADLOCK	103.0 ± 11.99 m	37.4 ± 19.15 m*	68.4 ± 31.58 m*	85.86 ± 9.31 m*
(% different)	(baseline)	(-63.69%)*	(-33.59%)*	(-16.64%)*
b) Computational Efficiency				
Time/Iter. (learning)	0.16 ± 0.04 ms	N/A	N/A	44.78 ± 18.94 ms
Time Comp. (learning)	$O(N)$	N/A	N/A	$O(E \cdot U \cdot N \cdot H^2)$
Time/Iter. (deployed)	0.16 ± 0.04 ms	0.11 ± 0.04 ms	0.11 ± 0.03 ms	18.13 ± 10.58 ms
Time Comp. (deployed)	$O(N)$	$O(N)$	$O(N)$	$O(N \cdot H^2)$

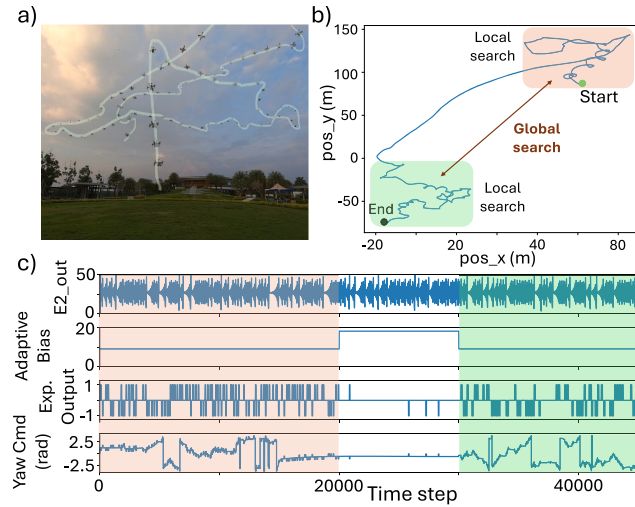


Fig. 10. (a) Low-angle time-lapse snapshots of the flying robot's exploration behavior (flying trajectory); a video can be seen at <https://www.manoonpong.com/ANEC/Video1.mp4>. (b) A top-view of the flying robot's exploration behavior (flying trajectory) shows both local and global search strategies; video demonstrating these exploration behaviors can be seen at <https://www.manoonpong.com/ANEC/Video2.mp4>. (c) The computational signals illustrate how the system processes data and guides the flying robot's exploration, with the highlighted colors (orange and green) corresponding to the exploration behavior indicated by the same colors in (b).

4.5. Real-world demonstrations

Real-world adaptive exploration scenarios were demonstrated by flying the robot outdoors, both in open areas and in environments with obstacles. Figure 10 shows the adaptive exploration behavior (flight trajectory) of the robot as it flew over an open area, accompanied by time-lapse snapshots. The flight trajectory exhibited both local and global search behaviors. Additionally, the computational signals are presented to illustrate how the system processes data and guides the robot's exploration. However, due to limitations in the flight time, communication range between the robot and ground control station, and the remote control and other online monitoring systems, the flying area was restricted to the area shown in Fig. 5 (approximately 12,000 m², or a 120 m diameter). To ensure that the robot remained within the active communication range and within our line of sight, the maximum flight velocity was fixed at a low value, i.e., 0.5 m/s. Manual control was used to move the robot back into the designated area if it flew beyond the limit.

Furthermore, we demonstrated the robot's adaptive behavior in an obstacle-filled area, where the control system enabled the robot to per-

form safe navigation and exploration. The results in Fig. 11(a) show the robot's flight trajectory as it avoids obstacles (O1, O2, O3, and O4), arranged as depicted in Fig. 11(b) and dummyTXdummy- captured from a camera (indicated in blue) positioned in Fig. 11(a). The green points (P1, P2, P3, and P5) indicate observation points from the back view of the robot, as shown in Fig. 11(c)–(f), respectively. Fig. 11(g) presents related signals, including normalized inputs from the LiDAR (right and left) where -1 means no obstacle in the range and 1 means that an obstacle is near (about 1 m distance), an obstacle avoidance output (A3 in ANOAC), exploration output (E6 in ANEC), yaw control command, and the actual yaw. The pink-highlighted area represents behavior driven by the ANOAC in relation to the pink dashed circle in Fig. 11(a), while the blue-highlighted areas represent behavior driven by the ANEC.

5. Discussion and conclusion

In this study, we proposed an adaptive neural control system that integrated two distinct neural dynamics of modular two-neuron recurrent networks for adaptive exploration and obstacle avoidance controls

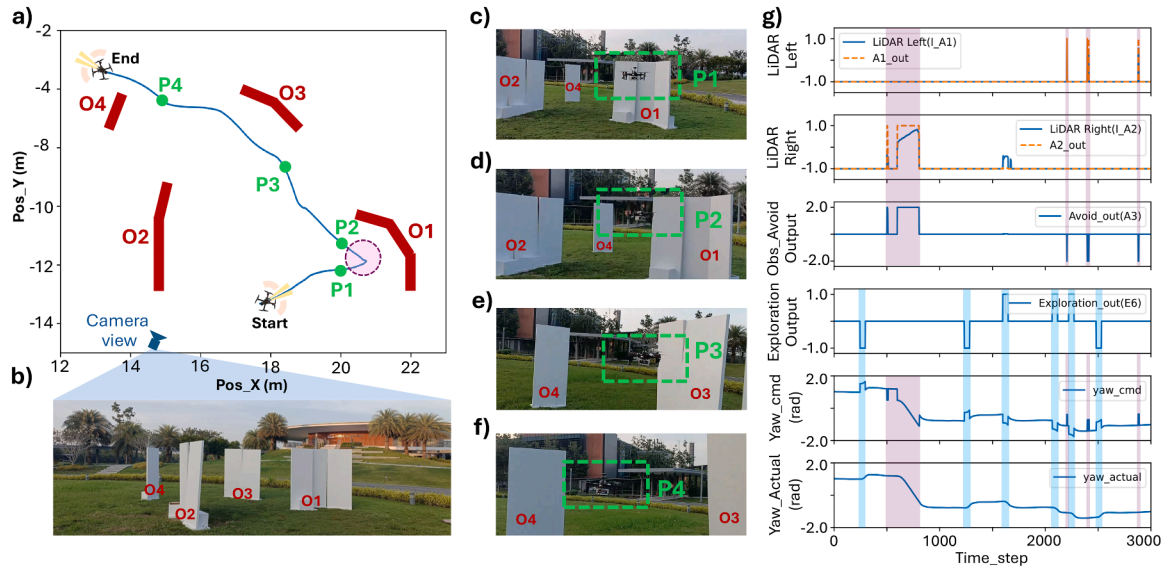


Fig. 11. (a) Top view of the flying robot's flight trajectory shows exploration behavior in an obstacle-filled area. (b) Arrangement of obstacles (O1, O2, O3, and O4). (c), (d), (e), and (f) Snapshot back views of the flying robot at green observation points (P1, P2, P3, and P4) in (a). (g) Related signals during the mission, including inputs from the LiDAR (right and left), obstacle avoidance output, exploration output, yaw control command, and actual yaw, while the pink-highlighted area represents behavior driven by the adaptive obstacle avoidance control in relation to the pink dashed circle in (a), and the blue-highlighted areas represent behavior driven by the adaptive exploration system. A video demonstrating this exploration behavior can be seen at <https://www.manoonpong.com/ANEC/Video3.mp4>.

of autonomous flying robot systems. Firstly, Lorenz attractor-like neural dynamics was embedded in the NLCN, which is a part of the ANEC. This dynamics drove the NLCN output into a Lorenz attractor-like pattern, which was subsequently processed by the post-processing network. This network employs an adaptive bias thresholding mechanism to generate the final adaptive exploration control commands (see Fig. 3). Secondly, hysteresis neural dynamics, which was formulated through the recurrent weights of neurons in the NHCN—a part of the ANOAC. The synaptic weights in the NHCN, consisting of recurrent and inhibitory weights, can be used to perform online learning through a synaptic plasticity mechanism. Two NHCNs were incorporated into the ANOAC, each corresponding to two front and two side-surrounding detection inputs. These NHCNs produced adaptive output commands to prevent collisions during exploration tasks (see Fig. 4). The integration technique applied to these two different neural dynamics involves processing frequency differentiation. The ANOAC, characterized by slow neurodynamics, operates at a normal frequency, while the ANEC, featuring faster neurodynamics, operates at a 50 times slower rate. This synchronization ensures that the flying robot effectively responds to both obstacle avoidance and exploration commands, aligning with the respective update frequencies of each system.

The adaptability and performance of the proposed neural control system were demonstrated by flying the simulated flying robot in an open environment. During the demonstration, the bias values were periodically adapted by alternately switching between 1.0σ and 2.0σ , with σ representing the standard deviation of the NLCN output. The results showed adaptive exploration behavior resembling both local and global search behaviors, alternately switching based on the adapted bias values. Furthermore, the overall analyzed flying trajectory, when compared with the mathematical Lévy flight (power-law) model, demonstrated significant similarity with the approximate power-law (Lévy) exponents (α) of 1.5. Moreover, we could observe behaviors when flying in closed environments measuring 100 m^2 ($10\text{ m} \times 10\text{ m}$). The experiments applied three different bias values (1.0σ , 1.5σ , and 2.0σ). The recorded flying trajectories revealed that adjusting the bias values results in varying search and exploration behaviors of the flying robot. When the bias value was set low (1.0σ), the flying robot frequently changed direction, leading to a circular flight pattern that predominantly focused on spe-

cific areas. This behavior can be characterized as a form of local search. Conversely, when the bias was set to higher values (1.5σ and 2.0σ), the flying robot exhibited fewer turns and traveled more in a forward direction, showing wall-following behavior in the given closed environments. This integration enhanced exploration performance. Specifically, it increased the flying distance by 69.98%, 91.54%, and 53.15% in environments characterized by low, high, and very high densities of obstacles with corners and deadlocks, respectively. In particular, the synaptic plasticity learning dynamics of this integrated system produced interesting results. The indirect influence of the ANEC system positively contributed to smoothing the learning process of synaptic plasticity in the ANOAC system. Furthermore, a performance comparison with other state-of-the-art exploration strategies, including Lévy flight, random walk, and DRL, showed that the proposed adaptive exploration method achieved better performance in both exploration distance and computational efficiency. Lastly, the implementation of the proposed neural control system was also demonstrated on a real flying robot.

These results effectively demonstrate the functionality, adaptability, and performance of our proposed adaptive exploration method across various flying robot platforms, provide comparisons with the mathematical Lévy flight (power-law) model, and achieve better performance than other state-of-the-art exploration methods. Here, we further discuss our proposed method in terms of the system properties and opportunities for further study.

5.1. Biobehavioral foundations of Lévy flight

In animal foraging behavior studies, biologists and ecologists analyzed foraging trajectories (Benhamou, 2007; Reynolds & Rhodes, 2009; Sims et al., 2008; Viswanathan et al., 1996), typically dividing them into two components: movement length and turning angle. However, in most studies, the primary focus of foraging behavior analysis was on investigating the distribution of the moving step length, while the distribution of the turning angles was often assumed to be random. Based on this analytical concept, foraging trajectories are classified as Lévy flight if the moving step length distribution follows a Lévy distribution or some other Lévy-like model. Conversely, the trajectories are classified as Brownian motion if the moving step length distribution resembles a

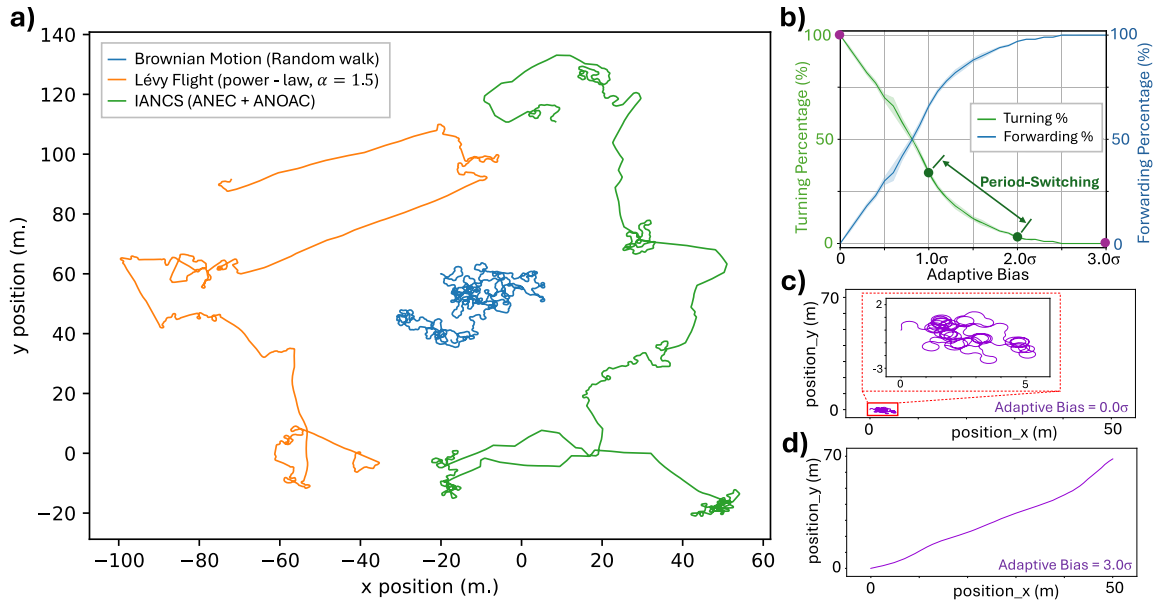


Fig. 12. (a) Comparison of the flying trajectories among the three exploration control strategies: Brownian motion or random walk (depicted in blue), Lévy flight (depicted in orange), and the proposed IANCS (depicted in green). (b) Analysis of the adaptive bias shows the proportion of turning and forward movements at different bias values. (c) Exploration behavior under a very small bias value of 0.0σ results in cyclic turning within a confined area. (d) Exploration behavior under a very large bias value of 3.0σ results in prolonged forward movement.

Gaussian or uniform distribution. Both Lévy flight and Brownian motion patterns have been observed in animal foraging behavior. For example, the tracked flying paths of wandering and black-browed albatrosses (Humphries et al., 2012) in their quest for fish exhibited both Lévy flight and Brownian motion patterns. Brownian motion was employed during prey capture, indicating a more focused and random search strategy, while Lévy flight was observed when resources were sparse and information was lacking, suggesting a more exploratory approach under such conditions. Moreover, some studies investigated the exploratory behavior of *Drosophila* larvae from a neuroscientific perspective. These studies involved allowing the larvae to perform exploratory behavior (i.e., moving freely in experimental arenas), both with and without blocked synaptic activity in the brain, subesophageal ganglion (SOG), and sensory neurons, to determine whether the larvae's exploratory behavior are intrinsically generated or attributable to sensory stimulation and environmental interactions. The results showed that the larvae's exploratory behavior persisted even when the brain, SOG, and sensory neurons were blocked (Berni, 2015; Berni et al., 2012). This suggests the existence of an autonomous exploratory behavioral program within the abdominal and/or thoracic networks of the ventral nerve cord (VNC), capable of operating independently of the brain and SOG and utilizing its intrinsic neural dynamics within the networks to generate complex exploratory movement patterns. When considering the overall movement trajectories, these movements exhibit Lévy-like patterns (Sim et al., 2019). However, when the nervous system (brain, SOG, and sensory system) was functioning normally, the larvae's crawling behavior showed further adaptability to the quality and distribution of food resources (Wosniack et al., 2022). Furthermore, in studies examining the movement of ants navigating through a given space, the results confirmed that ants exhibit lateral oscillations while in motion (Clement et al., 2023; Deeti et al., 2023; Grob & Jundi, 2023). These oscillations, combined with variations in forward speed, effectively optimize the distance covered while also enabling them to turn left and right simultaneously. Nevertheless, an effective ant navigation system also relies on the visual system and other sensory information (Buehlmann et al., 2020; Knaden & Graham, 2016; Sun et al., 2020, 2021).

Based on the biobehavioral studies mentioned above, we concluded that the underlying neural system driving exploratory and foraging be-

haviors in animals involves both internally generated neural dynamics and externally driven stimuli. These two systems work in coordination to generate optimal search and exploration behaviors during animal foraging. This study primarily focused on developing an adaptive neural system with an autonomous exploratory behavioral program. We achieved this by exploiting its intrinsic neural dynamics and incorporating an adaptive bias thresholding mechanism to generate adaptive exploration behavior using a simple predefined bias input. However, future investigations could explore the use of external stimuli to modulate the bias input of this thresholding mechanism. Figure 12(a) shows the comparison of exploration behaviors, and presents the trajectories generated by Brownian motion (random walk), Lévy flight (power-law, $\alpha = 1.5$), and the proposed IANCS, all within the same map. This visualization highlights the differences among the exploration strategies. Specifically, with the Brownian motion strategy (blue), the drone performed more looping flight patterns, concentrating its exploration within a relatively small area. In contrast, the Lévy flight (orange) and the proposed adaptive exploration (green) strategies produced longer forward flight patterns, enabling the drone to cover larger areas while occasionally focusing on smaller regions for exploration. Accordingly, we can clearly observe different exploration behaviors among Brownian motion (random walk, blue), Lévy flight (orange), and the proposed IANCS (adaptive exploration, green). Furthermore, the flight patterns of Lévy flight (orange) and the proposed IANCS (green) are also comparable.

The exploration behavior generated by the proposed IANCS, as shown in Fig. 12(a) (green), was achieved by dynamically adapting the bias value through a period-adapting strategy implemented via the adaptive bias thresholding mechanism in the ANEC. This process periodically switches the system's bias input values between 1.0σ and 2.0σ . The adaptive bias input directly influences the exploration behavior. For example, when using 1.0σ , the system produces more turning commands, causing the robot to turn frequently and exhibit local search behavior. In contrast, when 2.0σ is applied, the system generates fewer turning commands, allowing the robot to move forward for extended periods, thereby enabling global search behavior. Given this relationship between the adaptive bias input and resulting exploration behavior, the exploration behavior can be primarily controlled by adjusting

the adaptive bias input value. To further illustrate the impact of adaptive bias on behavior modulation, an analysis is presented in Fig. 12(b). This graph shows the percentage of turning and forward movements corresponding to different adaptive bias value settings in the thresholding mechanism, highlighting the control system's capability to modulate exploration behavior by adapting the bias input. Additionally, we demonstrated the robot's behavior under small and large bias values (i.e., 0.0 and 3.0) to illustrate the behavioral boundaries of the system. A bias value of 0.0 resulted in cyclic turning behavior within a confined area, as shown in Fig. 12(c), highlighted in the small red block and further detailed in the inset red dashed box. In contrast, a bias value of 3.0 led to prolonged forward movement, as illustrated in Fig. 12(d). A key feature of the proposed IANCS system is its ability to dynamically switch between local and global search strategies in real time by simply adapting the bias input precisely at the right time and place. This adaptability can be further extended into a sensor-driven adaptation strategy, enabling the system to perform local search to collect nearby food or global search to explore new resource areas. In this study, however, this capability was simplified to a predefined period-adapting strategy. In contrast, the traditional Lévy flight model regulates exploratory behavior by adjusting the Lévy exponent (α), a distribution-based parameter that shapes the probability distribution from which movement step lengths are drawn. Although real-time adaptation of the Lévy exponent (α) can influence overall exploration behavior, it does so indirectly by modifying the statistical properties of movement, rather than enabling immediate, context-specific behavioral adaptation. Therefore, the traditional Lévy flight model lacks the capability for dynamic, real-time control over search strategies, making it less adaptable than the proposed bias-driven mechanism.

5.2. Neurobiological basis of Lévy flight

In insect models, the neural basis of Lévy flight strategies involves the interaction of multiple neural systems supporting internal pattern generation and external sensory input processing (Buehlmann et al., 2020; Clement et al., 2023; Webb & Wystrach, 2016). This pattern generator, found in the VNC (Berni, 2015; Berni et al., 2012), is unlike a central pattern generator (CPG) that typically governs rhythmic locomotion (Büschges & Ache, 0; Ijspeert, 2008), and produces intrinsic complex outputs. These intrinsic outputs serve as a ground state, driving the insect to perform navigation and exploratory behavior with Lévy-like patterns (i.e., random walks with variable step lengths). Furthermore, these primitive patterns are modulated by external sensory cues (Buehlmann et al., 2020; Clement et al., 2023), enabling context-appropriate turning patterns for navigation across diverse environments. For example, biological studies have revealed that the lateral accessory lobes (LALs) (Namiki & Kanzaki, 2016; Steinbeck et al., 2020; Webb & Wystrach, 2016) contain a neural system that processes external sensory information to control turning patterns for obstacle avoidance and target approaching. While the precise neural integration and interaction mechanisms of the VNC pattern generator and LALs turning control remain unclear, our study here introduces a possible mechanism of their integration to produce final control outputs from a neural computation aspect. This integration enables safe and efficient navigation and exploration that complements goal-directed behaviors such as locating food, seeking shelter, or finding mates.

In biological systems, chaotic dynamics has been proposed as a mechanism underlying the emergence of Lévy flight search patterns in insects and other animals (Reynolds et al., 2016; Steinbeck et al., 2022). Inspired by this, we abstractly implemented chaotic dynamics derived from a Lorenz attractor within our NLCN. The network basically serves as an internal pattern generator which produces the ground-state pattern for Lévy flight search behavior. To facilitate the interaction between the internal generator's output and sensory inputs, we introduced an adaptive bias thresholding mechanism to selectively activate outputs that exceed the defined threshold. As a proof of concept, instead

of incorporating complex multimodal sensory cues to determine adaptive bias values, we first employed a simple rule that alternates the bias values among the standard deviation of the network's output. This enables the neural system to modulate exploratory behavior based on internally generated dynamics, implemented in our ANEC. For the LALs turning control, we utilized our NHCN (Grinke et al., 2015b; Pedersen & Manoonpong, 2018). It processes external sensory information and incorporates a synaptic plasticity mechanism (Citri & Malenka, 2008; Motanis et al., 2018) to produce adaptive, hysteresis-based dynamics for obstacle avoidance, implemented in our ANOAC. Finally, the outputs from the ANEC and ANOAC are combined through the IANCS to generate the final control commands, enabling Lévy flight search behavior with adaptive obstacle avoidance to navigate diverse environments with different obstacle densities.

5.3. Lévy flight implementation on exploration robots

The incorporation of Lévy flight in robotics has been extensively studied, with numerous works integrating the Lévy flight model into environmental exploration tasks as an internal pattern generator to replicate biological exploration behaviors (see Table 2). These studies leveraged the Lévy flight to guide robots in exploring large environments (Katada et al., 2022; Katada & Yamashita, 2022; Zedadra et al., 2018). Furthermore, the Lévy flight model for exploration has been combined with bio-inspired search algorithms, such as the firefly-based search (Zedadra et al., 2022) and beetle antennae search algorithm (Xin Xu & Shen, 2020), enabling robots to efficiently execute search and exploration tasks in foraging scenarios. In complex environments with obstacles, reactive obstacle avoidance (Zedadra et al., 2022, 2018) has been integrated to ensure safe and efficient exploration. Additionally, studies by Sardinha et al. (2020) and Alexis (2020) have explored how to efficiently adapt exploration behavior between local search and global search strategies.

However, most existing studies rely on Lévy models derived from mathematical formulas with Lévy distributions. While these models are effective at capturing movement patterns, they are often difficult to relate to the dynamics of biological neural systems. Although mathematical models are precise and useful for theoretical analysis, neural models offer significant advantages in terms of learnability and adaptability. By applying a neural model framework, we can explore and exploit various features such as network architecture, neural dynamics, synaptic plasticity, and learning mechanisms. Neural models also allow for integration and interactions with external stimuli through multiple sensory inputs (e.g., visual, olfactory, auditory, etc.), enabling more efficient and effective exploration and foraging behaviors. Furthermore, implementing complete exploration control in mathematical systems often requires an additional random component to generate movement directions for robots, which involves employing multiple generative systems. In contrast, this study proposes a neural model with a modular architecture for generating adaptive exploration that represents Lévy flight foraging behavior. Our model uses a single generative system, providing a unified exploration output command (encompassing both step length and direction) to produce adaptive exploration behavior. Additionally, the neural dynamics and interactions of our model can be monitored and visualized in real time using neural visualization tools such as NeuroVis (Srisuchinnawong et al., 2021). Furthermore, our proposed method was applied to other terrestrial robot platforms (i.e., a simulated legged robot and a real wheeled robot) using the same control parameters to demonstrate the generalizability and transferability of our adaptive neural control system. Interestingly, the proposed IANCS system was able to operate efficiently on resource-constrained hardware (ESP32 microcontroller), providing effective exploration and obstacle avoidance with real-time online learning, enabling adaptive responses to complex environmental conditions. The results are presented in the Supplementary Material, Section 11.

Table 2

Comparative analysis of the exploration control mechanism, obstacle avoidance control mechanism, and robot behavior. The exploration control mechanism was evaluated in two aspects: 1) the model used (mathematical model or neural model) to generate the Lévy flight exploration behavior, and 2) the type of internal pattern generator required to execute the exploration mission. A “multiple” pattern generator indicates that separate generators are needed: one for generating the moving step length and another for generating the heading direction. A “single” pattern generator indicates that only one generator is used to produce both the moving step length and heading direction. The obstacle avoidance control mechanism was assessed in terms of the control method employed. The robot behavior was compared across three dimensions: 1) the ability to perform both global and local search behaviors, 2) ability to avoid obstacles, and 3) ability to perform target search or approach. Note that “extendable” refers to the system’s potential for further development (for more details, refer to the Supplementary Material, Section 10).

Method	Exploration control mechanism		Obstacle avoidance control mechanism	Robot behavior		
	Lévy flight model	Internal pattern generator		Obstacle avoidance	Global&local search	Target search/approaching
Zedadra et al. (2018)	MI	Multiple	RI	✓	✓	✓
Katada et al. (2022), Katada and Yamashita (2022)	MI	Multiple	–	✗	✓	✓
Zedadra et al. (2022)	MI	Multiple	RI	✓	✓	✓
Xin Xu and Shen (2020)	MI	Multiple	–	✗	✓	✗
Sardinha et al. (2020)	MI	Multiple	–	✗	✓	✗
Alexis (2020)	MI	Multiple	–	✗	✓	✗
Ours	ANI	Single	ANI With synaptic plasticity	✓	✓	Extendable

MI: Mathematical Implementation, RI: Reactive Implementation, ANI: Adaptive Neural Implementation.

5.4. Proposed mechanism and outcomes

The proposed ANEC employs a neural system method to generate Lorenz attractor-like dynamics, demonstrating its capability to produce a complex, dynamic output signal. This generated signal is subsequently processed through a post-processing network equipped with an adaptive bias thresholding mechanism. This mechanism plays a critical role in enabling the system to generate adaptive exploration commands. Notably, this single system is capable of producing various movement patterns that encode two distinct Lévy movement mechanisms: one governing the step length (or travel distance) and the other controlling the orientation (or direction). When the ANEC is applied to the flying robot, its exploration behavior alternates between both local and global search patterns. Furthermore, empirical data from the flying robot’s trajectory was compared to the Lévy flight (power-law) model, which has an approximate power exponent (α) of 1.5. The comparison was conducted using the KS test, as shown in Fig. 7. The results indicated no significant difference between the two distributions ($p > 0.05$), suggesting that the empirical flight data follows a distribution closely resembling the Lévy flight model. Additionally, the ANOAC, which incorporates hysteresis neural dynamics, was integrated to ensure the robot can explore environments containing obstacles safely and efficiently. This integrated neural control system contributed to a significant increase in the exploration distance when navigating obstacle-filled environments, by 69.98 %, 91.54 %, and 53.15 % in environments characterized by low obstacle density, high obstacle density, and very high obstacle density with deadlock, respectively (see Fig. 8). The coordination of control commands between exploration and avoidance commands enables the flying robot to execute more significant turns, allowing it to escape stuck situations faster than when relying solely on ANOAC. This enhanced coordination allows the robot to move through complex obstacles more efficiently and explore greater distances. Furthermore, the exploration behaviors facilitated by the ANEC also indirectly contribute to the smoothing of the learning process in the ANOAC, as illustrated in Fig. 9.

5.5. Further enhancement direction

The proposed ANEC methodology opens up promising avenues for the future research aimed at investigating alternative neural model architectures and the dynamics of synaptic weights, expanding the potential beyond traditional Lévy flight models. This methodology is underpinned by two key components: the NLCN and adaptive bias thresholding mechanism. The NLCN autonomously generates output signals

that exhibit Lorenz attractor-like dynamics, contributing to complex, chaotic behavior. However, beyond these attractor dynamics, the network also demonstrates other neural dynamics, including periodic oscillations. These oscillatory patterns can be modulated by adjusting network parameters such as synaptic weights, thereby enhancing the flexibility of the system. The adaptive bias thresholding mechanism is implemented through a period-adapting method. This approach involves switching between different values of σ (the system’s bias input), such as alternating between 1.0σ and 2.0σ . This simple yet effective adjustment increases the adaptability of the control system, allowing it to dynamically respond to changes in the environment and exploration needs. Looking forward, there are significant opportunities to explore the properties and dynamics of these two key components in greater depth. For example, studying the neural dynamics of the NLCN could involve experimenting with changes in network parameters, such as variations in transfer functions, synaptic weights, inputs, and bias values. This may provide insights into how different neural configurations impact the system’s ability to generate adaptive exploration behavior (Table 2). Moreover, future work should focus on developing and integrating multisensory mechanisms (Buehlmann et al., 2020; Freas & Spetch, 2023; Liu & Borisjuk, 2024) to adapt neural dynamics and bias values. Incorporating external stimuli, such as visual cues or other sensory inputs, could enhance the system’s adaptability and responsiveness to its environment. For instance, we plan to integrate sensory feedback, along with adaptive trajectory planning based on dynamic movement primitives with online temporal scaling adaptation (Rothomphiat et al., 2024) (see the Supplementary Material, Section 12), for adaptive disturbance compensation. This integration could lead to a more robust and versatile system capable of handling diverse real-world conditions and compensating for external disturbances like wind speed changes. Additionally, investigating the collaborative dynamics between the intrinsic neural generator (the NLCN) and external sensory inputs, as well as the impact of external disturbances, could clarify the mechanisms driving effective search and exploration behaviors. This may offer valuable insights into the biological principles behind efficient search and foraging strategies in nature, particularly in the context of food acquisition (Klages, 2023; Sun et al., 2023; Wosniack et al., 2022).

To enhance the capabilities of ANOAC, further improvements can be made so that the flying robot can adaptively and robustly handle complex, highly cluttered, or dynamically changing environments (e.g., environments with small and moving obstacles and trees). This can be achieved by integrating advanced, high-dimensional sensors, such as a depth camera and multi-directional LiDAR, along with a sensory processing module to pre-process complex input data before transmitting

the data to the ANOAC. Furthermore, for both safety and efficient navigation, speed control becomes a critical component. In scenarios where the autonomous drone or robot operates at high speeds, it is essential for the autonomous drone or robot to dynamically adapt or reduce its speed in order to avoid potential collisions. To address this, an online learning-based speed adaptation control module (Jaiton et al., 2022) can be integrated into our modular neural control system. This enhancement will enable the flying robot to safely and effectively explore target areas with varying environmental complexity by adapting its flying speed online.

In the future, we plan to enhance the proposed neural control system by exploring how the intrinsic neural generative system collaborates with visual and sensory inputs. By leveraging the NLCN and adaptive bias thresholding strategies, this investigation will guide development and refinement of the neural control system. Additionally, we intend to further improve the ANOAC module to enable the system to operate effectively in more complex and challenging environments, including dynamic and cluttered obstacle-rich scenarios. We aim to integrate the insights gained from these studies into the neuro-adaptive search and exploration control system, enhancing the creation of more effective foraging robots. Our research will continue to build on the principles of neurorobotics (Jaiton & Manoonpong, 2023; Ramdya & Ijspeert, 2023), focusing on behavioral mechanisms that can shed light on how animals navigate and interact with their environments during foraging tasks. Ultimately, we aim to contribute to a deeper understanding of the neurobiological underpinnings of efficient foraging strategies and advanced robotic systems for autonomous exploration and navigation in complex environments.

CRediT authorship contribution statement

Vatsanai Jaiton: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Poramate Manoonpong:** Writing – review & editing, Validation, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the startup grant on Bio-Inspired Robotics (P.M., Principal Investigator) from Vidyasirimedhi Institute of Science and Technology (VISTEC) and Siam Commercial Bank (SCB). We thank Dhamdhawach Horsuwan and Matas Manawakul for their support with real-world robot testing.

Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.neunet.2025.107913](https://doi.org/10.1016/j.neunet.2025.107913).

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